

# How Tinted Are Your Glasses? Gender Views, Beliefs and Recommendations in Hiring\*

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This draft: April 6, 2026

## Abstract

We study gender gaps in a multi-stage hiring process, focusing on recommendation and recruitment. Across two experiments—one with academic economists and one with a college-educated sample—we show how recommendation letters can cause gender hiring gaps. While recruiters favor female candidates, consistent with efforts to diversify hiring, some groups of recommenders choose less effective letters for women, undermining this advantage. Strategic but erroneous beliefs about letter effectiveness and gender views of recommenders explain these choices. A two-sample prediction exercise using letters from the academic job market supports the experimental finding that gender views drive the language used in recommendations.

*JEL Codes:* A11, C93, D90, J16.

*Keywords:* Diversity, Experiments, Gender, Recruitment.

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\*Certified random order of the authors ([Ray and Robson, 2018](#)). We thank participants at the Nottingham CeDEX brown bag seminar, the BEE lab talk at the University of Michigan, the Bristol Applied Economics Meetings, the RES conference in Birmingham, and the WeRise workshop in Boston University, seminars at the Norwegian School of Economics, and the FAIR & Max Planck Workshop in Bergen for constructive comments and suggestions. We have benefited greatly from discussions with and inputs from Malte Baader, Björn Bartling, Marianne Bertrand, Leonardo Bursztyn, Stefano DellaVigna, Nickolas Gagnon, Alex Imas, Rafael Jiménez-Durán, Lawrence Katz, Patrick Kline, John List, Katrine Løken, Paula Onuchic, Amanda Pallais, Evan Rose, Anya Samek, Sofia Shchukina, and Marie-Claire Villeval. We are grateful to Alex Imas for sharing relevant experimental material from [Bohren et al. \(2022\)](#). We also thank Cristina Griffa and Yuliet Verbel for their research assistance. We gratefully acknowledge funding from the University of Nottingham. The usual disclaimers apply. Corresponding author: [fabio.tufano@leicester.ac.uk](mailto:fabio.tufano@leicester.ac.uk).

# 1 Introduction

Despite progress in educational attainment and labor force participation, gender gaps in hiring remain persistent—particularly in highly skilled occupations (e.g., [Bertrand et al., 2010](#); [Goldin et al., 2017](#); [Tesar, 2025](#)). These disparities raise important questions about how hiring decisions are made and which features of recruitment processes contribute to gender inequality. A key challenge is that recruitment typically involves multiple stages and actors, including recommenders and hiring committees. This complexity can obscure the mechanisms through which disparities arise, making it difficult to identify where and how gender gaps emerge. Understanding how different stages of the recruitment process shape hiring outcomes is therefore essential, both to uncover the sources of inequality and to inform the design of effective policies.

In this paper, we focus on two central stages of recruitment: the recommendation stage, in which candidates receive a letter characterizing their strengths, and the hiring stage, in which recruiters evaluate candidates based on these letters and other information. Recommendation letters are a standard component of selection procedures in high-skilled occupations—including academia, law, medicine, and international organizations (e.g., [Avery et al., 2001](#); [Roth, 1984](#); [Coles et al., 2010](#))—and have also been shown to improve employment prospects in lower-skilled labor markets (e.g., [Abel et al., 2020](#); [Heller and Kessler, 2024](#)). Because recommendation letters shape how candidate quality is communicated to recruiters, they represent a potentially important channel through which gender disparities may arise.

Existing work has largely modeled recommendations as a binary signal—either provided or not (e.g., [Pallais and Sands, 2016](#); [Abel et al., 2020](#)). The content of recommendation letters, however, varies along multiple dimensions. A growing literature has examined the language used in such letters, documenting systematic gender differences in how candidates are described in academic labor markets. In particular, women are more frequently portrayed using ‘grindstone’ terms emphasizing diligence, and less frequently using ‘ability’ terms emphasizing cognitive skills (e.g., [Trix and Psenka, 2003](#); [Schmader et al., 2007](#); [Madera et al., 2009](#); [Eberhardt et al., 2023](#); [Baltrunaite et al., 2025](#)). These differences also correlate with hiring outcomes. However, existing studies are primarily observational and cannot distinguish between alternative mechanisms, such as unobserved candidate characteristics, recommenders’ strategic responses to perceived recruiter preferences, or discrimination.

To make progress in addressing these issues, we conduct two complementary, pre-registered experimental studies. The first is a survey experiment with academic economists that closely mirrors the institutional environment of the junior academic job market — a setting where recommendation letters carry an important weight in the hiring process and where gendered language has been documented in previous research. We recruit over 1,000 economists from research-intensive institutions and randomly assign them to the role of recommenders or recruiters. Recommenders select one of three letters for a hypothetical candidate—a letter emphasizing ability, one emphasizing grindstone traits, or a neutral letter—and report their beliefs about the effectiveness of each one. Recruiters evaluate the same candidate, whose qualifications are held constant apart from randomly assigned

gender, and indicate which letter would make them most likely to interview the candidate. We also classify participants according to their familiarity with research on gender differences in recruitment — a measure we use throughout the paper to distinguish between informed and uninformed gender views. This design allows us to isolate the causal effects of candidate gender and letter content, and to examine the mechanisms underlying recommendation and hiring decisions in a realistic professional context.

To complement this setting and assess the broader applicability of our findings, we conduct a second, incentivized online experiment with more than 2,000 participants on Prolific. This experiment simulates a complete labor market in which participants are assigned to the roles of candidates, recommenders, or recruiters. Candidates complete general knowledge quizzes that provide measures of productivity, cognitive ability, and effort. Recommenders observe partial performance information, choose a letter type, and report beliefs about the effectiveness of each letter. Recruiters select among candidates based on their letters, with payoffs tied to the selected candidate’s unobserved performance. Finally, for both recommenders and recruiters, we assess to what extent they hold stereotyped gender views on effort and ability. This design serves two purposes. First, it allows us to study recommendation and hiring decisions in a controlled environment where candidate quality is observable to the experimenter, enabling us to assess how recommendation letters distort the transmission of information. Second, it extends the analysis beyond academia to a broader, college-educated population, allowing us to evaluate whether the mechanisms identified in the academic labor market generalize to other contexts.

Across both studies, we find that gender plays distinct roles at the recommendation and hiring stages. At the hiring stage, recruiters favor women over equally qualified men, consistent with efforts to increase diversity in the workplace. Recruiters also place substantial weight on letters emphasizing ability, making the choice of letter type important for a candidate’s prospects.

At the recommendation stage, however, some recommenders are less likely to choose ability letters for women, compared to equally qualified men, and these gender differences in recommendation letters can attenuate the intended diversification in hiring outcomes. We identify recommenders’ gender views and strategic but erroneous beliefs as key determinants of gendered letter choices. First, those with uninformed or stereotyped views are less likely to choose ability letters for women, while the other group does the opposite. These patterns are confirmed in a two-sample prediction exercise, in which we re-analyze actual reference letters from the academic job market for economists collected by [Eberhardt et al. \(2023\)](#). Second, recommenders typically select the letter they believe maximizes hiring prospects, highlighting the importance of strategic considerations. However, they systematically underestimate the effectiveness of ability letters for women relative to men. These beliefs are closely associated with participants’ gender views and are consistent with motivated reasoning (e.g., [Bénabou, 2015](#); [Stötzer and Zimmermann, 2024](#)).

To quantify how recommendation and hiring stages jointly shape gender disparities, we apply the Kitagawa–Oaxaca–Blinder decomposition ([Kitagawa, 1955](#); [Oaxaca, 1973](#); [Blinder, 1973](#)) to separate gender differences in hiring into a direct and a systemic component (see [Bohren et al., 2022](#); [Baron et](#)

al., 2024). The direct component captures differential treatment of male and female candidates with identical signals, while the systemic component captures disparities arising from gender differences in the signals themselves. We find that the direct component favors women: conditional on identical recommendation letters, women are more likely to be hired than men. However, this advantage is undermined—and in some cases offset—by the systemic component. Because women may be less likely to receive ability letters, and recruiters do not fully account for these differences in signal generation, gender disparities in recommendations can undermine diversification in hiring.

These findings have important implications. Much of the existing policy focus, including Diversity, Equity and Inclusion (DEI) initiatives, has targeted discrimination at the hiring stage. Our results suggest that such interventions, while potentially effective, may be insufficient to eliminate gender gaps. A substantial share of disparities arises earlier in the recruitment process, through gendered differences in how candidate quality is communicated. Moreover, these differences are driven in part by recommenders’ inaccurate beliefs about the effectiveness of different signals. Interventions aimed at correcting these beliefs—for example, by providing information about how recommendation letters influence hiring decisions—may therefore represent a promising and underexplored avenue for reducing gender disparities in recruitment (e.g., Boring and Philippe, 2021).

**Related Literature and Our Contribution.** Existing research has extensively documented the presence of discrimination in recruitment, both through field data (e.g., Goldin and Rouse, 2000; Riach and Rich, 2002) and correspondence studies (e.g., Bertrand and Mullainathan, 2004; Jacquemet and Yannelis, 2012). Recent studies have also explored the role played by recommendations in driving these differences. Although recommendations can improve efficiency, they can exacerbate gender gaps if men and women differ in their likelihood of being recommended (on efficiency, see, Pallais and Sands 2016; Dustmann et al. 2016; Abel et al. 2020; on gender gaps, see Beaman et al. 2018). Our paper contributes to this literature by investigating the actual content of recommendations rather than focusing only on whether a recommendation was sent or not. Contemporaneous research investigates this theoretically and, in an application, uses Large Language Models to illustrate the impact of gendered recommendations on hiring (Bohren et al., 2025). In contrast, we combine two experiments that analyze reference messages chosen by a population of academics and college-educated individuals and further relate them to observational data from the academic job market. This setup enables us to explore mechanisms in detail, in particular the role of gender views and strategic considerations.

Exploring such mechanisms links us to a body of work that has investigated behavioral channels underlying discrimination in hiring. Specifically, studies have examined the role of idiosyncratic preferences (e.g., Oreopoulos, 2011; Kuhn and Shen, 2013; Weber and Zulehner, 2014), evaluator’s gender (e.g., Bagues and Esteve-Volart, 2010; Deschamps, 2023), incentives (e.g., Szymanski, 2000; Parsons et al., 2011), uncertainty (e.g., Hendricks et al., 2003), and beliefs (e.g., Glover et al., 2017; Barron et al., 2024). Our paper contributes to this literature by showing how gender views and strategic considerations—consistent with motivated beliefs (e.g., Eyting, 2022; Stötzer and Zimmermann, 2024)—drive hiring gaps. We show that recommenders who believe they are acting in the

candidate’s best interest may unintentionally harm their prospects. Additionally, we extend recent research on discrimination in layered systems, which has developed empirical strategies to identify discrimination arising at different stages of a process (e.g., [Kline et al., 2022](#); [Bohren et al., 2022](#); [Baron et al., 2024](#); [Jassal, 2024](#); [Shaffer and Harrington, 2024](#)). In particular, our analysis uncovers the importance of gender views and strategic considerations on the *recommendation* rather than the *recruitment* side in explaining differential treatment.

Finally, our paper offers novel insights into the role of gatekeepers in explaining gender disparities in academia, especially in math-intensive subjects such as economics (e.g., [Ceci and Williams, 2009](#); [Breda and Ly, 2015](#); [Sarsons, 2017](#); [Lundberg and Stearns, 2019](#); [Hengel, 2022](#)).<sup>1</sup> Previous studies have used observational data to document gendered patterns in reference letters (e.g., [Correll et al. 2020](#); [Eberhardt et al. 2023](#); [Baltrunaite et al. 2025](#)). Our experimental results provide causal evidence on the role of gendered recommendations in recruitment outcomes. Moreover, by highlighting the importance of recommenders’ strategic considerations, our work points to the role of diffusing information about the efficacy of different types of recommendations as a policy option to diversify the profession.

The remainder of this paper is structured as follows. Section 2 outlines the design of our two experimental studies. Section 3 reports descriptive statistics. Sections 4 and 5 provide results separately for recruiters’ hiring decisions and recommenders’ letter choices. As part of the recommender analysis, we explore the role of gender views in letter choice both in our experimental data and in a re-analysis of observational data from the Economic Job market. Section 6 discusses additional mechanisms, while Section 7 brings both sides of the market together to evaluate the overall effect on hiring. Section 8 concludes.

## 2 Design and Procedures

We conduct two studies to isolate (i) the causal effect of recommendation letters on hiring outcomes, (ii) the effect of the candidate’s gender on the type of letter they receive, and (iii) how they interact in shaping gender gaps in hiring. The studies allow us to shed light on the mechanisms behind the effects, namely the role played by gender views, inherent differences between male and female candidates, and strategic considerations (e.g., whether recommenders choose the letter they believe maximizes their candidate’s success).

The first study relies on a novel survey experiment carried out with a large sample of academic economists. The second consists of an online experiment featuring a labor market with college-educated participants recruited on Prolific. Both studies received ethical approval from the Nottingham School of Economics Research Ethics Committee in May 2022.

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<sup>1</sup>Although women’s representation in economics has stagnated in recent years (e.g., [Tesar, 2025](#)), there are occasional signs of change as, for instance, positive gender gaps in economics fellowships and awards (e.g., [Card et al., 2022, 2023](#)).

## 2.1 Academic Survey Experiment

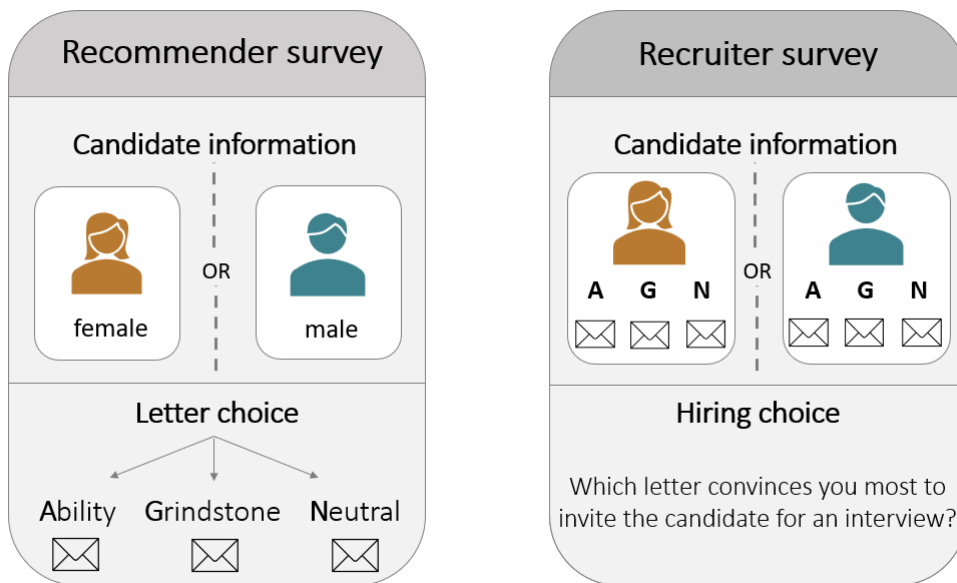
We designed the academic survey experiment (*‘academic survey’*) to capture the relevant features of the junior job market for academic economists. In a nutshell, academics participating in the study are either assigned to choose recommendation letters for hypothetical PhD candidates or assigned to review a hypothetical job market application package and decide whether to invite the candidate for an interview.

### 2.1.1 Design

Figure 1 presents a schematic overview of the survey structure. All instructions can be found in Appendix B.2. Participants are randomly assigned to either the *recommender survey* (left panel) or the *recruiter survey* (right panel). All participants see a short biography of one hypothetical candidate including information about their name, PhD institution, job market paper, and research field. As standard in this literature, the gender of the candidate is varied *between* subjects to reduce experimental bias (e.g., [Bertrand and Duflo, 2017](#)), and is signaled to participants through the first name of the candidate (e.g., [Lippens et al., 2023](#)). To avoid any connotations beyond gender, we ran a pre-survey to select names and chose *Julia* and *Thomas*; both names were rated as unequivocally female and male, respectively, but not perceived differently with respect to ethnic and socioeconomic background.

In the **recommender survey**, participants are shown one candidate (either female or male) and asked to select one of three summary recommendation texts for the candidate. The order in which participants view the recommendation texts is randomized between participants. These texts are

Figure 1: Survey structure



*Note:* We vary the candidate's gender (as implied by their first name) between subjects. Recommenders choose one of three letters to recommend the candidate. Recruiters see all three letters and decide which one convinces them most to invite the candidate for an interview.



written to mimic the final paragraphs of recommendation letters, which usually summarize the overall view of the recommender on the candidate’s strength and potential (see Appendix B.2.2 for the exact phrasing of the paragraphs). Focusing on the summary recommendation allows us to keep the survey manageable (the text is about 140 words).

The summary paragraphs are identical except for three sentences that are either neutral, stress the candidate’s natural ability (e.g., ‘bright and creative’), or emphasize grindstone attributes (e.g., ‘extremely hardworking’). We selected these words based on common ‘ability’ or ‘grindstone’ attributes in previous analyses of reference letters in the economic job market (Eberhardt et al., 2023). All participants had the option to view the full, three-page version of each letter (see Appendix B.3). However, only 12% of participants did so, and not differentially for male or female candidates.

Next, we elicit beliefs on which letter is perceived to be the best by other participants in the study. This question aims to capture recommenders’ strategic considerations when choosing a letter. In particular, we ask which letter maximizes a candidate’s chance of being invited to an interview. The beliefs question is incentivized; if recommenders choose the letter that is preferred by most other participants, they increase their chance of winning a voucher (see Section 2.1.2 for more details on incentives).

Note that to enhance task engagement and experimental control, our design abstracts from reputational concerns that real-world recommenders may face in repeated interactions. While this is a limitation of our design with respect to external validity, it provides at the same time a valuable simplification, as it allows us to capture key behavioral drivers and identify the role of beliefs and biases in a highly controlled setting.

In the **recruiter survey**, participants receive the profile of one candidate (either female or male) along with all three letters (i.e., grindstone, ability, or neutral). Again, the order in which letters are presented is randomized, and recruiters could click to access the full letter. Recruiters were more likely than recommenders to access the full letter, but again only a minority decided to do so (27%). Recruiters then choose which one out of the three letters would most likely lead them to invite the candidate for an interview.<sup>2</sup> We chose to ask about individual preferences for invitations to interviews instead of final hiring prospects to abstract from departmental politics and other complexities in the hiring process. For simplicity, in the rest of the paper, we still refer to the recruiters’ choice as their ‘hiring decision’.

For all participants (recommenders and recruiters), we collect additional information, including demographics (i.e., gender, academic seniority, year of PhD, the country and the RePEc ranking of their institution), previous experience of writing reference letters and recruiting in the economics job market. In addition, to ascertain individual awareness of gender issues in academia, we ask about their familiarity with research on the topic. Specifically, participants self-report their knowledge of five recent academic papers on gender gaps in recruiting (Wu (2020); Dupas et al. (2021); Koffi (2021);

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<sup>2</sup>In addition, we ask recruiters how likely they would be to invite a candidate with a given letter on a scale from 0 to 10. The median score for each letter was 9 out of 10, indicating that our hypothetical candidates were perceived as very strong. Given the lack of variation in this measure, we focus on the relative hiring choice in our analysis.

Hengel (2022); Eberhardt et al. (2023)). We then construct a binary indicator based on a median split, distinguishing between participants who report knowing at least one of these papers (47%) and those who report knowing none (53%). We use this measure in our analysis as a proxy for the informedness of participants' gender views and test how the latter shape letter choices and hiring decisions. The reason we decided on this measure is that a sample of academics likely has some familiarity with standard measures of gender views which therefore may be particularly prone to desirability biases. Familiarity with specific research papers, by contrast, is less normatively loaded and thus a more neutral alternative for this sample. Our working assumption is that academics who know more about gender research are, on average, more interested in the topic and tend to hold more progressive gender views. We refer to those academic economists who know at least one paper as having 'informed' views and those who do not as having 'uninformed' views.

### 2.1.2 Procedures

Our target sample for the academic survey consists of academics that act as recommenders and/or recruiters for the job market.

To contact participants, we collected the email addresses of academic economists from publicly accessible institutional websites. We chose institutions that regularly engage with the international job market for economists, resulting in a population of 13,185 academics based in 21 countries (see Appendix B.1 for more details). We discuss possible differences between actual recommenders in the economics job market and the population of academic economists contacted for the survey in Section 3.

The survey was programmed in Qualtrics (2005), pre-registered (AsPredicted #101261), and conducted in July 2022 with a median response time of 6.8 minutes.<sup>3</sup> To incentivize participation, we donated £2 to UNICEF for each completed survey and respondents who finished the study received a lottery ticket to win one of fifty £50 Amazon vouchers. If they answered the beliefs question correctly, they received an additional 5 tickets. Every lottery ticket was equally likely to be drawn. In total, we received 1,020 complete responses (506 in the recommender and 514 in the recruiter survey), which is in line with our pre-specified target sample of 1,000 participants.

## 2.2 Online Experiment

The academic survey allows us to observe experts' behavior on both sides of the job market in a controlled environment. As references are widely used in labor markets (particularly, for highly skilled professions), a natural question is whether the patterns observed hold for a broader population facing choices that are not hypothetical, but have immediate monetary consequences.

To answer this question, we designed an online experiment that complements the academic survey in several ways. First, it provides monetary incentives for all decisions. Second, it strengthens

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<sup>3</sup>All regressions presented in this paper are robust to dropping the fastest and slowest 5(10)% of responses. If anything, effects are estimated more precisely when winsorizing on response time. To conserve space, we do not report these robustness checks in the appendix. Results are available upon request.



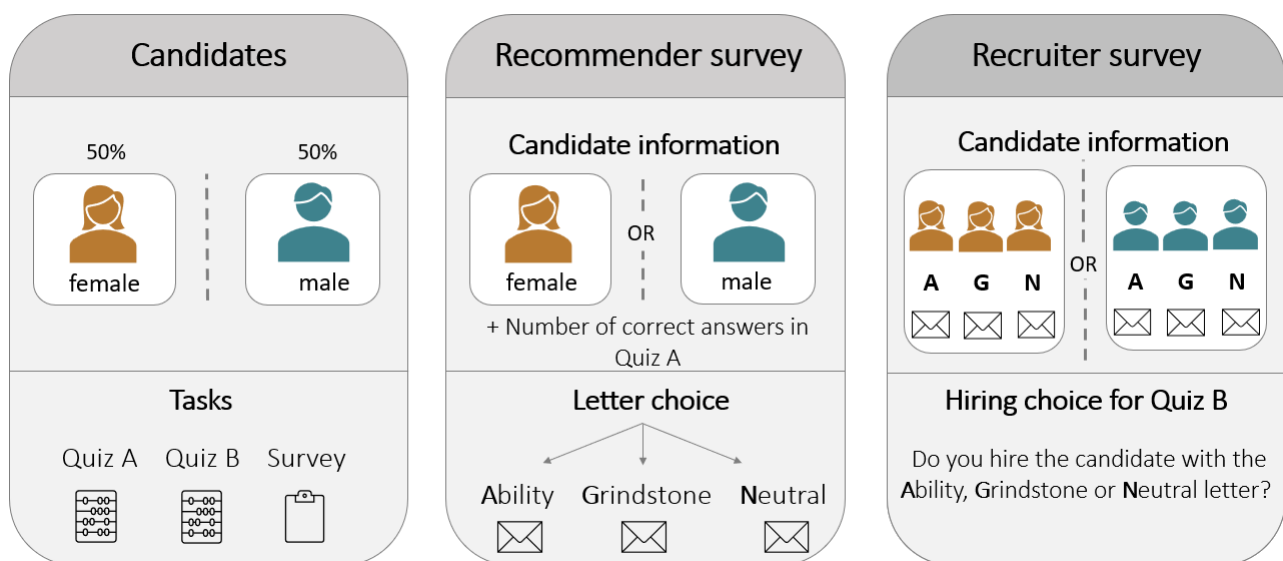
experimental control, since we can elicit candidates’ actual productivity, cognitive skills, and self-reported effort. Third, it moves beyond the specific sample of academic economists. The experiment was conducted on the online platform Prolific.

### 2.2.1 Design

In the experiment, participants are part of an online labor market and are assigned to three different roles: *candidates*, *recommenders*, and *recruiters*. The setup builds on [Bohren et al. \(2022\)](#) and is designed to study the effects of job recommendations. The basic structure is as follows: Candidates complete a range of tasks that generate an imperfect signal of their productivity. Recruiters want to hire candidates for a task, but do not know their productivity. Recommenders observe an informative signal and send a recommendation message — which we call ‘letter’ — to recruiters. We again use a between-subjects design to minimize experimenter demand effects: recommenders and recruiters are shown information for either female or male candidates. We provide a more complete description of participants’ incentives and choices below, while Figure 2 offers an overview of the design of the online experiment. All instructions can be found in Appendix B.4.

**Candidates** have to solve two quizzes: Quiz A and B. Each quiz consists of ten incentivized questions about finance, mathematics and history. To reduce the number of performance levels to evaluate, we focus on candidates who correctly answered either 5, 6, or 7 Quiz-A questions (13% of all candidates). Given that the average number of correctly solved questions was 3.48, these are high-performing individuals (ranked between the 86<sup>th</sup> and 99<sup>th</sup> percentile). We restricted the sample to high-performing candidates to align the quality of candidates with our highly positive recommendation letters and to reflect the context of recruitment for high-skilled positions. Both recommenders and recruiters were informed about the average performance of 3.48, so our chosen performance levels provide a

Figure 2: Experimental design



*Note:* We vary the candidate’s gender between subjects. Recommenders see one of three performance levels (5, 6, or 7 correctly answered questions). Recruiters see all three letters and have to decide which letter convinces them most to hire the candidate.

salient quality signal without appearing unrealistic.

As additional controls for candidate quality, we collect proxies for grindstone and ability characteristics among candidates. In particular, we survey participants on how much effort they put into the quiz and elicit cognitive skills using a standard Cognitive Reflection Test ('CRT', [Thomson and Oppenheimer, 2016](#)). Although our effort measure is self-reported — and thus imperfect — we find that it significantly correlates with time spent on the quizzes (the correlation coefficient is  $r = 0.46$ ,  $p < 0.001$ ).

**Recommenders** observe the performance of one candidate only in Quiz A, as measured by the number of correctly answered questions, along with their gender. While a high correlation between quizzes is likely, recommenders are thus not perfectly informed about the potential of a candidate. First, half of all recommenders are assigned a woman, the other half a man. Next, for each assignment, recommenders are equally likely to be assigned a candidate who answered 5, 6, or 7 questions correctly. They are then asked to choose one of three letters (highlighting again grindstone, ability, or neutral attributes) to recommend the candidate to recruiters. The order in which letters are presented is randomized between participants. The exact phrasing of the letters can be found in Appendix [B.4.2](#). If the candidate is hired, the recommender receives a bonus (see Section [2.2.2](#) for details on incentives). Recommenders thus earn a monetary reward for sending the strongest possible letter. As in the academic survey, we also elicit strategic considerations, that is beliefs about which letter is maximizing the candidate's chance to be hired.

**Recruiters** are shown three different candidates (as indicated by different candidate IDs). The three candidates have the same gender, but received different letters. Recruiters lack any information about the quiz performance and can thus only rely on the letter and/or gender to make a hiring decision. While they do not know the exact performance, all three letters are overall very positive and strongly recommend the candidate. Recruiters are incentivized to choose the best candidate, as their bonus depends on how many questions the candidate answered correctly in Quiz B.<sup>4</sup>

In sum, the experiment follows a 2x3 design for both recommenders (male/female candidate x 5/6/7 correct answers in Quiz A) and recruiters (male/female candidate x ability/grindstone/neutral letter). In addition, for both recommenders and recruiters, we collect a measure of their gender views. In line with research in social psychology on gender brilliance bias (e.g., [Del Pinal et al., 2017](#); [Bian et al., 2018](#); [Storage et al., 2020](#)), we ask them if they believe that men/women are successful due to effort, ability or a combination of both. In particular, we ask participants "Do you think that male/female workers are characterized a) more by talent, b) more by effort, or c) equally by effort and talent?" If a participant chooses talent for male workers/effort for female workers, we characterize them as having stereotyped gender views (28% of our sample). Otherwise, we classify them as having

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<sup>4</sup>In line with the academic survey and common recruiting decisions, we focus on the relative hiring choice, in which recruiters have to make a choice among several candidates. In addition, we also elicit an absolute hiring decision: recruiters see a single candidate and decide whether to hire them. If they hire the candidate, their bonus depends on the candidate's performance and ranges from £0 to £1; if they do not hire the candidate, they receive a fixed payment of £0.40.

non-stereotyped views (72% of our sample).<sup>5</sup> This allows us to explore the role of gender views as a mechanism for explaining both differences in letter choice and hiring decisions between male and female candidates.

Finally, we collect data on how both recommenders and recruiters perceive the candidate, the skills required to perform well in the quizzes, their demographic background (i.e., age, gender, education), their self-reported risk (Dohmen et al., 2011), and strategic sophistication which we measured through a beauty contest game (Nagel, 1995). Demographics as well as perceptions and preferences are used as additional controls when examining letter choice and hiring decisions.

### 2.2.2 Procedures

The experiment was programmed in Qualtrics (2005) and pre-registered (AsPredicted #102500). Participants were recruited via Prolific in July 2022. We restricted the study to participants in the US and the UK to minimize language barriers. To enhance comparability with the academic sample and due to the experiment involving hiring decisions, we required participants to have at least an undergraduate degree. We collected data for 100 candidates, as well as 959 recommenders and 959 recruiters. We then match several recommenders to one candidate and we do so in a way that ensures balance across candidates' gender and performance levels.

All decisions and elicited beliefs were incentivized. Candidates received £0.10 for each correctly solved quiz question and CRT question. Recommenders received a bonus of £0.50 if their candidate was hired, plus respectively £0.50 if their beliefs about the recruiters and the candidate were correct. Recruiters received £0.10 for each Quiz B question their hired candidate solved correctly plus £0.50 if their beliefs about the candidate were correct. In addition, participants received a show-up fee, resulting in average earnings of £2.06.<sup>6</sup> As participants took on average 7.2 minutes to complete the experiment, this is equivalent to an hourly wage of £17.20.<sup>7</sup>

## 2.3 Hypotheses

Our experimental design allows us to explore the causal effect of recommendation content on hiring decisions and the mechanisms underlying gendered language in recommendation letters. Our key focus is thereby on comparing grindstone to ability letters. Neutral letters are included in the design to make this comparison less salient to participants.

As for the hiring decision, we expect the language used in the recommendation letter to matter for candidates' prospects.

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<sup>5</sup>Recruiters are directly asked their own opinions. Recommenders are asked what opinion they think most recruiters have. Appendix Figure A.2 shows that answers are very similar. Note that the group of non-stereotyped participants includes 80%, who think men and women are equally characterized by talent and effort, and 20% with reversed views who choose effort for men or talent for women. The behavior of those two sub-groups is qualitatively identical. For this reason and because the second group is very small, we do not distinguish between the two in our analysis.

<sup>6</sup>The average payments by roles were £2.73 for workers, £2.14 for recommenders and £1.91 for recruiters.

<sup>7</sup>Again, all regressions are robust to winsorizing on response time and dropping the fastest and slowest 5(10)% of responses.

**Hypothesis 1:** *Grindstone letters lower the probability of being hired relative to ability letters for both male and female candidates. Not receiving an ability letter is thus costly for candidates.*

Next, we turn to letter choices. Building on research by [Eberhardt et al. \(2023\)](#) on the junior academic job market for economists, we expect recommendation language to be gendered. In particular, we expect men and women to be described differently by recommenders, disadvantaging women.

**Hypothesis 2:** *There is a gender bias in letter choice. Compared to male candidates, female candidates are less likely to be described with an ability letter.*

Finally, we expect strategic considerations to shape the choice of recommendation letter. We expect recommenders to select the letter they believe maximizes a candidate’s chances of being hired, even if these beliefs may be incorrect.

**Hypothesis 3:** *Gendered letter choices are driven by strategic considerations.*

The academic survey and the online experiment are designed to tackle the same research questions but complement each other in several respects. The academic survey closely links to existing observational evidence on gendered letter choice in the academic job market. By targeting a similar population while experimentally ruling out gender differences in unobserved characteristics, we can speak to the mechanisms underlying previous empirical findings. The online experiment, by contrast, broadens the analysis to a less specialised participant pool and offers even greater experimenter control, as all decisions are incentivized and we can provide a fully objective performance measure for candidates. Given that the two experiments share the same research questions and hypotheses, we present results for both studies in parallel throughout the paper.<sup>8</sup>

### 3 Descriptive Statistics

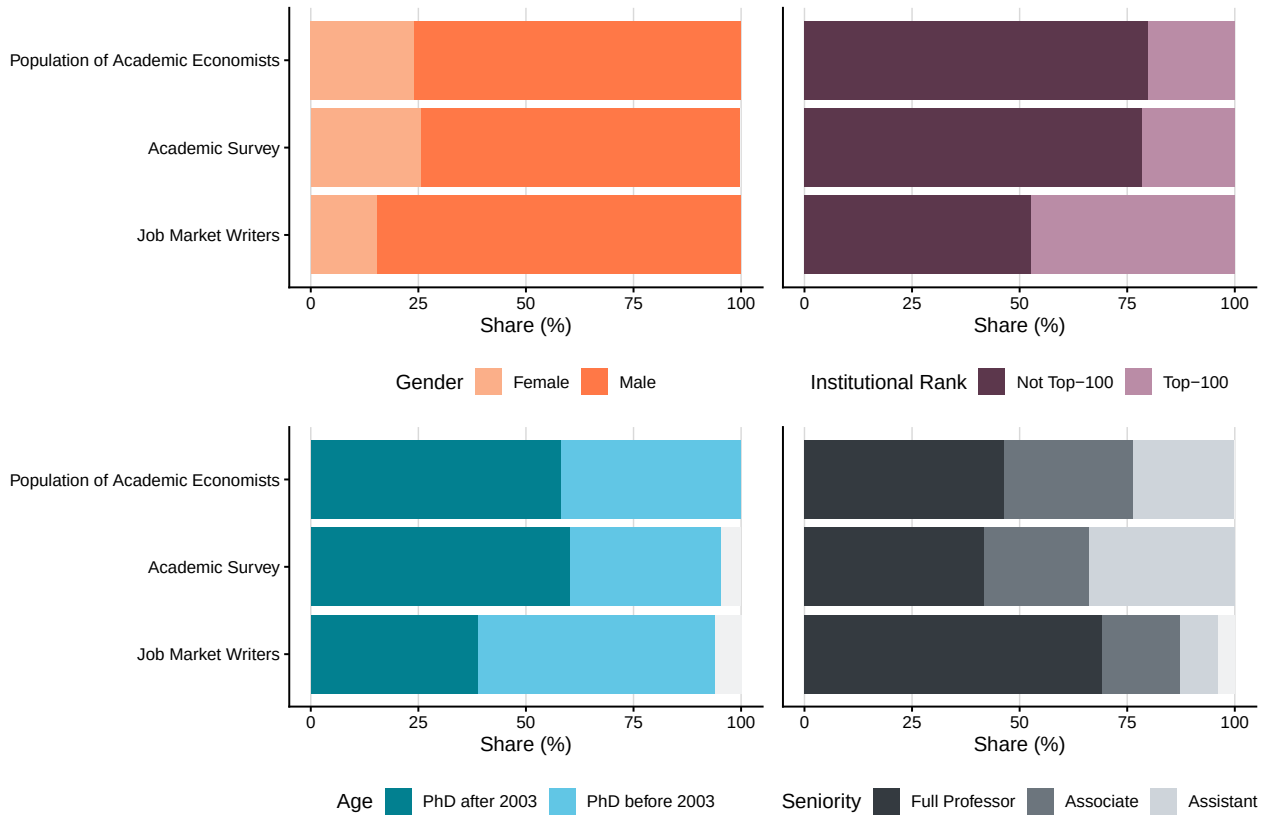
Before turning to our findings, we describe the demographic composition of the samples in the experiments. For comparison, we also report the characteristics of actual Economics job market recommenders in the large sample compiled by [Eberhardt et al. \(2023\)](#) (*JM* data). This sample is constructed from approximately 3,000 applications for assistant professor positions received by a UK research-intensive institution during the 2017-2021 period.

Figure 3 presents summary demographic characteristics for three groups relevant to the academic survey: (i) the population of academic economists, defined as all economists invited to participate in the survey; (ii) the academic survey sample, consisting of those who participated; and (iii) job market recommenders, comprising academic economists in the *JM* data. The figure shows that the population of academic economists and survey respondents are highly comparable across observable characteristics: in both groups, women account for roughly 25%, approximately 20% are affiliated with top-100 institutions, and close to 60% received their PhD after 2003. The distributions of academic rank are also similar, with modest differences across groups — assistant professors comprise

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<sup>8</sup>These hypotheses have been formulated in our pre-registration plan ([AsPredicted #101261](#)), where we also include potential heterogeneous effects along a number of different individual characteristics.

Figure 3: Demographic characteristics across samples



**Notes:** The figure shows the demographic composition across three samples of (potential) letter writers used in this paper. The population of academic economists refers to the entire group of academic economists invited to participate in the academic survey. The academic survey refers to the sample of academic economists who participated in the study. Job market recommenders are the academic economists writing the recommendation letters present in the observational sample collected by [Eberhardt et al. \(2023\)](#).

23% of the population and 34% of respondents, associate professors 29% and 24%, and full professors 46% and 41%, respectively.

In contrast, the sample of recommenders in the *JM* data is different from the population of academic economists and thus our survey sample. Recommenders are less likely to be women (15% among recommenders versus 24% in the population), work at more prestigious institutions (49% versus 20%), and hold more senior positions (72% versus 46%). These differences are to be expected, since academic advisors tend to be more senior, and PhD candidates may reach out to academics in more prestigious institutions to write their letters. To account for the fact that the survey sample is not representative of *JM* recommenders, we weight the survey sample in all analyses.

Participants in the online experiment are drawn from a different population that does not consist of academic economists but is still college-educated (see Appendix Table A.2 for demographic characteristics). The experimental sample is thus different enough to gauge the generalizability of our findings, but similar enough to draw a meaningful comparison to the academic survey. Besides profession, a striking difference with the academic sample is that we have a balanced gender composition. Contrasting to the academic survey data, the online experiment was not designed to mimic the academic job market. Consequently, we do not expect recommenders in the online experiment

to be similar to recommenders in the  $\mathcal{JM}$  data and do not weight the experimental sample in our main analysis. In the appendix, we confirm the robustness of all our results to weighted regressions based on the gender composition of recommenders in the  $\mathcal{JM}$  data.

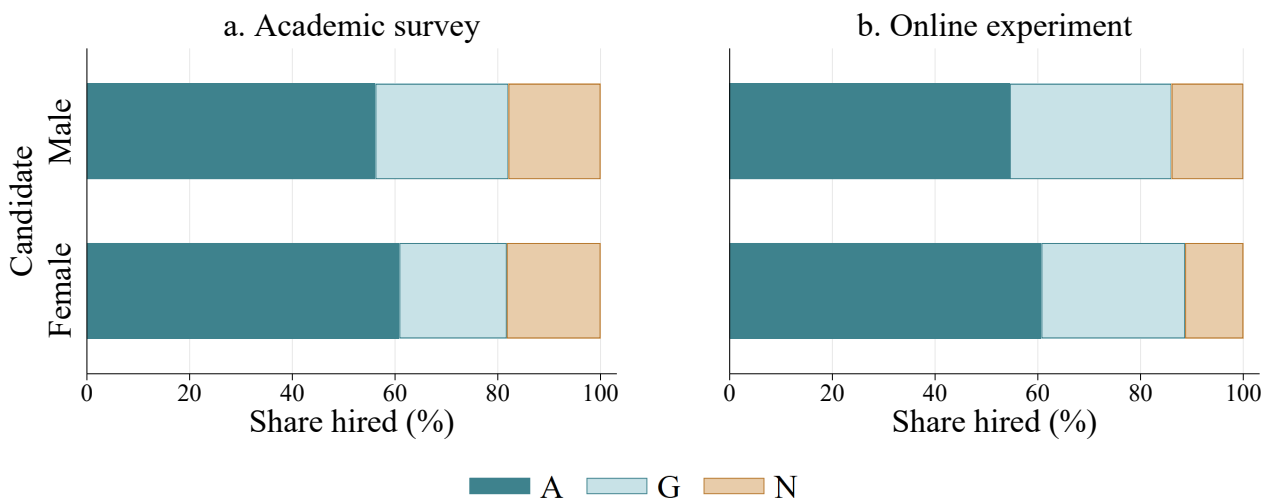
## 4 Hiring Decisions

In this section, we document that letter type matters for hiring decisions and that the same letter affects recruitment outcomes differently for men and women. Here, we abstract from the fact that men and women may receive different letters. We focus on gender differences in recommendation letters in Section 5.

To investigate hiring decisions in the recruiter survey, we use a multinomial logit regression model where the dependent variable equals 0 if the candidate with the neutral letter was hired, 1 if the candidate with the grindstone letter was hired, and 2 if the candidate with the ability letter was hired. Our explanatory variable of interest is a gender indicator and the corresponding marginal effect estimates the predicted probability of a woman being hired relative to a man with the same letter type. Our analysis controls for order effects (i.e., the order in which reference letters were presented to subjects) and recruiter characteristics.

Figure 4 plots the raw data on hiring decisions. It shows that recruiters in both studies have a clear preference for candidates with an ability letter, confirming Hypothesis 1. Grindstone and neutral letters, by contrast, are much less effective. This pattern is particularly pronounced for female candidates. In the academic survey (online experiment), the share of women hired with an ability letter is approximately 40 (33) p.p. higher than the share hired with a grindstone letter. For men, the corresponding difference is only 30 (23) p.p. This highlights that women benefit disproportionately

Figure 4: Hiring choices in the academic survey and online experiment



*Note:* Share of candidates hired with an ability (A), grindstone (G) or neutral (N) letter. In the academic survey, recruiters decide which letter makes them most likely to invite a given candidate to an interview. In the online experiment, recruiters make a hiring choice between three candidates of the same gender but with different letters.

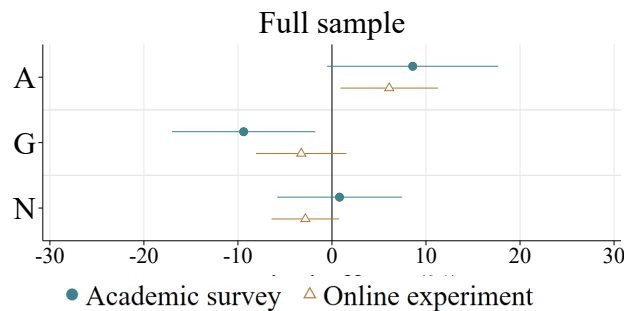


from an ability letter compared to men.<sup>9</sup>

Next, we investigate the causal impact of gender on recruitment by letter type using the multinomial regression analysis described above. Figure 5 plots the marginal effects alongside 90% confidence intervals for female relative to male candidates. Compared to their male counterparts, female candidates are more likely to be hired with an ability letter (academic survey: 8.59 p.p.; online experiment: 6.09 p.p.), and less likely to be hired with a grindstone letter (survey: -9.39 p.p.; experiment: -3.27 p.p.). These differences are significant for ability letters in the online experiment and for grindstone letters in the academic survey. Neutral letters display no statistically significant gendered patterns. The gender gap in hiring is significantly different between ability and grindstone letters in the academic survey and the online experiment (Wald tests, survey:  $p = 0.06$ , experiment:  $p = 0.099$ ) and between ability and neutral letters in the online experiment ( $p = 0.05$ ).

Overall, these findings indicate that receiving an ability rather than a grindstone or neutral letter (see Figure 4) is beneficial for all candidates, but this effect is more pronounced for women (see Figure 5). Even though there are some differences between the two studies, the results point qualitatively in the same direction. Full regression results are provided in Appendix Tables A.3 and A.4.<sup>10</sup>

Figure 5: Differences in hiring between female and male candidates



*Note:* The figure plots marginal effects from multinomial logit regressions, where the dependent variable is 0 if the candidate with the neutral letter (N) was hired, 1 if the candidate with the grindstone letter (G) was hired, and 2 if the candidate with the ability letter (A) was hired. Regressions control for order effects and include recruiter controls. Whiskers represent 90% confidence intervals. See Appendix Tables A.3 and A.4 for full set of estimates. Results for the online experiment become even more pronounced when using weighted regressions (see Appendix Table A.5).

<sup>9</sup>In addition to the choice between candidates, we ask academic recruiters to rate how likely they are to invite a candidate with a given letter to an interview on a scale from 0 to 10. We find that women have an about 0.5 points higher rating independently of the letter (Wilcoxon rank-sum tests,  $z = -3.84$ ,  $p < 0.001$ ). This reflects a general propensity to favor women in (at least part of) the recruitment process. Similarly, recruiters in the online experiment are on average 8.56% more likely to hire a woman if asked whether they want to hire a given candidate without comparing them to others (Wilcoxon rank-sum test,  $z = -2.72$ ,  $p = 0.007$ ). On average, recruiters choose to hire in 61% of cases.

<sup>10</sup>When looking at recruiter controls, we find that in the academic survey, participants from the Top 100 institutions are more likely to hire a candidate with an ability letter ( $p = 0.06$ ). In the online experiment, female recruiters are more likely to hire candidates with a neutral ( $p = 0.07$ ) or a grindstone letter ( $p = 0.01$ ) compared to male recruiters. Risk preferences and strategic thinking play no role in hiring decisions.

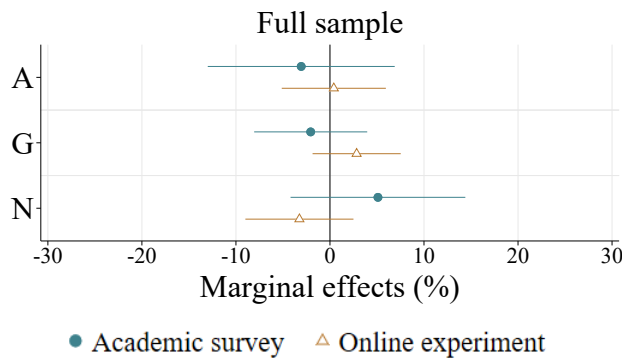
## 5 Letter Choices

The previous section showed that recruiters react differently to the same letter type depending on the candidate’s gender. By design, recruiters were equally likely to see each letter type for male and female candidates. However, in practice, recommenders may choose different letters for men and women. We now turn to study whether this is the case. In section 7, we bring both elements—letter choice and recruiters’ reaction to letters—together to evaluate the overall impact of gender and letters on hiring outcomes.

To investigate recommender decisions, we again use a multinomial logit regression model where the dependent variable is 0 if the recommender chose a neutral letter for the candidate, 1 if they chose the grindstone letter, and 2 if they chose the ability letter. Our variable of interest is the gender indicator, and the corresponding estimated marginal effect captures the predicted probability that a woman receives a given letter type relative to a man.

When examining the full sample of recommenders, we find no significant gender differences in the probability of receiving any letter type (see results in Figure 6). Taken at face value, this finding does not align with earlier results in the literature (e.g., [Eberhardt et al. 2023](#)). We consider two complementary explanations for this difference.

Figure 6: Difference in letters between female and male candidates



*Note:* The figure plots marginal effects from multinomial logit regressions, where the dependent variable is 0 if a neutral letter (N) was chosen, 1 if a grindstone letter (G) was chosen, and 2 if an ability letter (A) was chosen. Regressions control for order effects and include recommender controls. Whiskers represent 90% confidence intervals. See Appendix Tables [A.6](#) and [A.7](#) for the full set of estimates.

First, we examine a substantive explanation: aggregate effects may mask heterogeneity across groups. To analyze this possibility, we study the role of different gender views in the academic survey and online experiment in shaping letter choices.

Second, we leverage data from [Eberhardt et al. \(2023\)](#) to provide evidence on how sample composition and gender views drive gendered patterns in recommendation letters from the Economics Job Market. Both analyses are discussed below.

## 5.1 Heterogeneity across gender views

In this section, we explore whether gendered patterns in letter choices vary across sub-samples in our data. Specifically, in the spirit of existing theoretical contributions (e.g., [Onuchic, 2024](#)) and empirical studies that investigate mechanisms and sources of discrimination (e.g., [Fershtman and Gneezy, 2001](#); [List, 2004](#); [Neumark, 2018](#)), recommenders’ preconceptions about gender could influence their opinions about candidates, independently of the candidate’s quality.<sup>11</sup>

To investigate this possibility, we split the sample depending on gender views and find stark differences.<sup>12</sup> Figure 7 plots the marginal effects for the female dummy obtained from estimating the multinomial logit regression described above for different sub-samples (see Appendix Tables A.6 and A.7). For the online experiment, the regression includes the candidate’s quiz performance as a control, in addition to recommender controls. In both the academic survey and the online experiment, we uncover a similar pattern. Recommenders who have uninformed/stereotyped views are substantially less likely to give women an ability letter (-14.61 p.p. and -14.94 p.p. in the respective studies) and more likely to give them a neutral letter (14.63 p.p. and 10.37 p.p., respectively).<sup>13</sup> These effects are significant at the 10% level and robust to excluding recommender controls. Wald tests show that the gender gap for the ability letter is significantly different from the gender gap for the neutral letter (survey:  $p = 0.03$ , experiment:  $p = 0.03$ ) and in the case of the online experiment also from the one for the grindstone letter ( $p = 0.03$ ).

Our results thus show that recommenders with uninformed/stereotyped views behave more in line with the recommenders analysed in earlier observational studies (e.g. [Eberhardt et al. 2023](#)), providing qualified support for Hypothesis 2: consistent with our prediction, women are less likely to receive ability letters from this sub-sample of recommenders.

Among recommenders who have informed/non-stereotyped views, we instead see the exact opposite pattern. They are substantially more likely to give ability letters to female candidates (13.73 p.p. and 7.65 p.p.) and less likely to give them a neutral letter (-8.90 p.p. and -9.22 p.p., only significant for the online experiment). The gender gap for the ability letter in this sub-sample is significantly different from the gender gap for the grindstone letter (Wald tests, survey:  $p = 0.07$ , experiment:  $p = 0.099$ ) and in the case of the online experiment also from the one for the neutral letter ( $p = 0.02$ ).

Lastly, when interacting gender views with the candidate’s gender, Wald tests show that the gender gap in both the academic survey and the online experiment differs significantly across participants with different gender views for the ability letter (survey:  $p = 0.02$  and  $p = 0.001$  respectively) and the neutral letter (survey:  $p = 0.02$ , experiment:  $p = 0.005$ ). A joint Wald test of the interaction coefficients is also statistically significant (survey:  $p = 0.06$ , experiment:  $p = 0.005$ ), highlighting

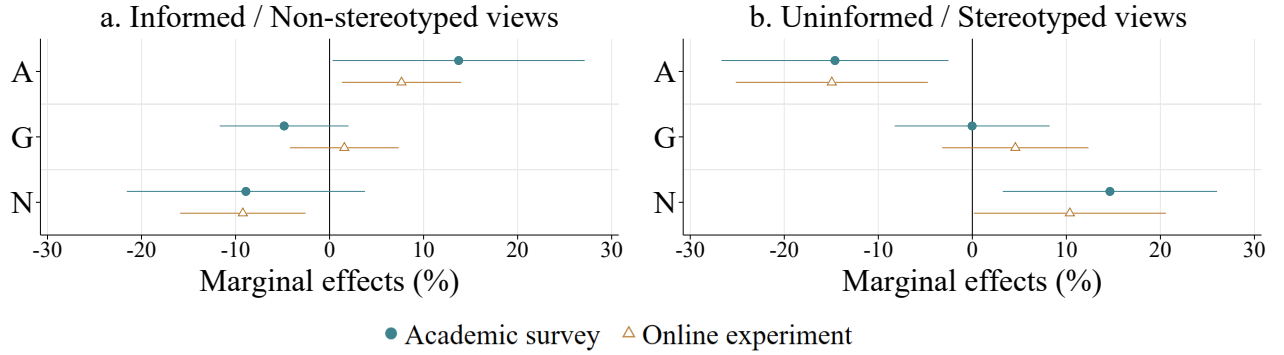
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<sup>11</sup>Our pre-registration anticipated potential heterogeneous effects across several individual characteristics, but it did not include a fully specified pre-analysis plan (e.g. [Imai et al., 2025](#)).

<sup>12</sup>We also analyzed *hiring* decisions in split samples but found that the same qualitative patterns hold in both groups – see Appendix Tables A.3 and A.4 for details.

<sup>13</sup>When looking at how other recommender controls affect letter choice, we find that in the academic survey, participants with more hiring experience are less likely to choose the grindstone letter ( $p = 0.006$ ). In the online experiment, participants with a higher level of education ( $p < 0.001$ ) and men ( $p = 0.02$ ) are more likely to send an ability letter. Risk preferences and strategic thinking play no role for letter decisions.

Figure 7: Differences in letters between female and male candidates



*Note:* The figure plots marginal effects from multinomial logit regressions, where the dependent variable is 0 if a neutral letter (N) was chosen, 1 if a grindstone letter (G) was chosen, and 2 if an ability letter (A) was chosen. Regressions control for order effects and include recommender controls. Whiskers represent 90% confidence intervals. See Appendix Tables A.6 and A.7 for the full set of estimates. Results for the online experiment become even more pronounced when using weighted regressions (see Appendix Table A.8).

that the effect of the candidate’s gender differs across gender views.

Overall, this heterogeneity contributes to explaining the absence of statistical significance in the aggregate results. Importantly, the convergence of patterns across our two experiments — each using different measures of gender views — supports the robustness and generalizability of the results. In the next section, we show using a two-sample prediction exercise that gender views also likely shape letter choices in the *JM* data. In line with an emphasis on cumulative evidence, we view our two experiments and this exercise as *conceptual replications* (e.g., Maniadis et al., 2017).<sup>14</sup>

## 5.2 Comparison to Economics Job Market (*JM*) data

To further examine these patterns, we turn to the data from Eberhardt et al. (2023), which contains applications for assistant professor positions submitted to a UK research-intensive institution between 2017 and 2021, to understand how our experimental findings relate to actual reference letters written for candidates on the Economics Job Market (*JM*). In particular, we ask whether the gendered patterns in letter choice are more pronounced among recommenders whose characteristics correlate with uninformed gender views in the academic survey.

To answer this question, we first note that in the *academic survey*, we know that men, those from earlier PhD cohorts, and with higher levels of seniority are also more likely to hold uninformed views (see Appendix Figure A.1). We then analyze the letters in the *JM* sample to ascertain the role of these characteristics on gendered patterns in letter choices. To mimic the survey experiment, we focus on applicants who graduated less than one year before going on the market. The sample covers almost 9,000 letters, written by over 4,000 recommenders for 2,900 candidates. To obtain a discrete categorization of letters akin to the one designed in the survey, we create mutually

<sup>14</sup>A conceptual replication uses a different method from the original approach to test the same underlying hypothesis. This corresponds to the third notion of replication discussed in Levitt and List (2009) and implemented by Alempaki et al. (2019). Consistent with Maniadis et al. (2017), convergent evidence across settings and replications can be especially informative, complementing pre-registration (e.g., Coffman and Niederle, 2015).

exclusive categories. Letters are classed as *Ability* (A), respectively *Grindstone* (G), if the emphasis on ability (grindstone) attributes is above the sample median.<sup>15</sup> This naturally leads to two further types: *Neutral* (N), containing neither attributes, and *Mixed* (M). We then replicate the same analysis implemented for the academic survey using the *JM* data. In particular, we run multinomial logit regressions, where the dependent variable is the type of letter a candidate receives (Ability, Grindstone, Neutral, or Mixed), and the variable of interest is an indicator equal to one if the candidate is female.

The upper panel in Figure 8 shows the results. Each marker in the plot represents the estimate for the female coefficient obtained from separate regressions run on the full sample as well as restricted sub-samples of recommenders, with darker shading for statistically significant coefficients. The estimates for the *full sample* show that overall, female candidates are significantly less likely to receive an ability letter and significantly more likely to receive a grindstone one (by 3.5 p.p. and 2.26 p.p., respectively). No significant gendered patterns emerge for mixed or neutral letters. These findings broadly replicate Eberhardt et al. (2023).

In addition, the upper panel in Figure 8 presents the regression estimates for 8 different sub-samples defined by the recommender’s gender (male vs. female), institutional rank (top-100 institution vs. other), academic rank (full/associate professors vs. other), and PhD cohort (early vs. late). When restricting the sample to male recommenders, female candidates are significantly less likely to receive ability letters and more likely to receive grindstone ones, whereas the effect is statistically insignificant for female recommenders. Likewise, recommenders in higher-ranked institutions exhibit more pronounced gendered patterns in their letters than those in lower-ranked ones, and the same holds when comparing senior to junior recommenders (defined by either academic rank or PhD cohort). Comparing these sub-samples to predictors of uninformed gender views in the academic survey, we find that the same demographics that correlate with *uninformed views* in the survey also drive gendered patterns in *JM* recommendation letters (see Appendix Figure A.1).

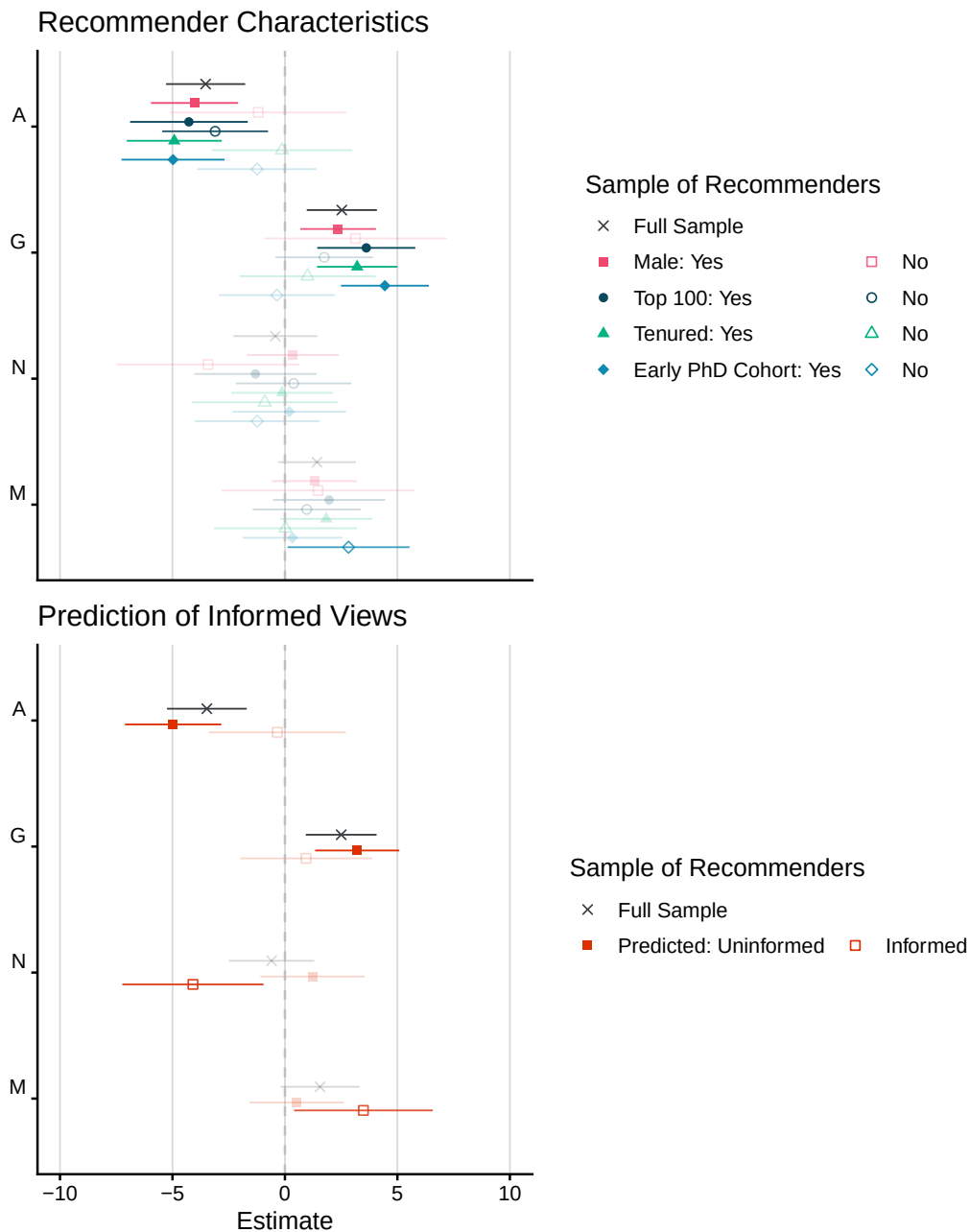
These descriptive splits are informative but they only consider each characteristic in isolation. We go one step further in assessing whether gender views can predict heterogeneity in the *JM* letter. Specifically, we build an informed-view propensity index using the survey and examine whether this prediction rule helps explaining gendered patterns in the *JM* letters.

Our empirical strategy follows a two-sample predictive approach: (i) we estimate the relationship between recommender characteristics and the informed-views index in the survey data; (ii) we then apply the estimated coefficients to the *JM* data to generate out-of-sample predictions of informed-views and classify recommenders accordingly; (iii) we analyze heterogeneity across the resulting strata. A similar strategy is applied in studies of teacher value-added, place effects, and college value-added (e.g., Chetty et al., 2014; Chetty and Hendren, 2018; Mountjoy and Hickman, 2021).<sup>16</sup>

<sup>15</sup>Appendix section A.4.1 presents the sample in more details and discusses the text analysis methods used to measure these attributes.

<sup>16</sup>In these papers, the index is estimated at the unit level (e.g., a teacher fixed-effects for value-added; county forecasts of causal place effects; college value-added) and then applied out-of-sample to classify units and study heterogeneity. Our setting differs only in the index construction: because ‘informed views’ are unobserved in the *JM* data,

Figure 8: Differences in letters received by male and female candidates in the *JM* sample – Split sample results according to recommender characteristics



**Notes:** The figure plots marginal effects from multinomial logit regressions, where the dependent variable is the choice of an ability (A), grindstone (G), neutral (N), or mixed (M) letter. Whiskers represent 90% confidence intervals. The top panel presents split-sample results according to the demographic characteristics of recommenders. The bottom panel presents split-sample results according to the variable predicting informed gender views (see main text). The sample is split based on whether the likelihood of holding informed views lies below or above 0.5. For ease of presentation, transparency is increased for non-significant results, see Appendix tables A.10 and A.11 for the full set of estimates.

Specifically, to generate the index, we use the *academic survey* sample to estimate a linear probability model for holding informed views, conditioning on respondent *i*'s gender, academic rank, PhD

we estimate a prediction rule in the survey and apply the fitted coefficients to *JM* recommender covariates to obtain a recommender-level predicted propensity – once transported into the *JM* data, this index is used in the same way as in the papers cited: that is, attached to units, employed for classification, and then used to analyze heterogeneity in outcomes.



cohort, and institutional rank (marginal effects are qualitatively identical if we use a logit model—results available on request):

$$\Pr(\text{Informed view}_i = 1 \mid X_i^{\text{AS}}) = \alpha + \beta X_i^{\text{AS}} + \varepsilon_i,$$

retain the estimated marginal effects  $\hat{\beta}$ , and treat them as a prediction rule. In the second step, we project this prediction rule onto the  $\mathcal{JM}$  recommenders’ demographics to compute, for each recommender  $j$ ,

$$\hat{p}_j^{\text{JM}} = \alpha + \hat{\beta} X_j^{\text{JM}}.$$

We classify recommenders according to whether their predicted propensity to hold informed views exceeds 0.5. Finally, in the third step, we explore heterogeneity in letter choices using this classification. The bottom panel of Figure 8 reports the resulting split-sample estimates. Recommenders predicted to hold uninformed views (propensity below 0.5) exhibit clear and significant gendered patterns in their letter choices, whereas those predicted to hold informed views display no statistically significant effects. Patterns are consistent across different cutoffs (see Appendix Table A.11).

The survey-based prediction of informed views thus successfully isolates the subset of  $\mathcal{JM}$  recommenders whose letters exhibit significant gendered patterns. This suggests that sample composition is key to reconciling the  $\mathcal{JM}$  evidence with the experimental findings:  $\mathcal{JM}$  recommenders most closely resemble survey respondents with uninformed gender views—the sub-group that makes gendered letter choices.

## 6 Further explanations for gendered letter choices

In the previous section, we showed that men and women receive different letters depending on the recommenders’ gender views. Beyond these views, letter choices could reflect two alternative mechanisms: First, men and women could genuinely differ in their capacity to work hard or in their innate ability. Second, letter choices may reflect strategic considerations, with recommenders believing that certain types of information are more beneficial for women than for men. We examine each of these alternative explanations in turn below.

### 6.1 Are male and female candidates different?

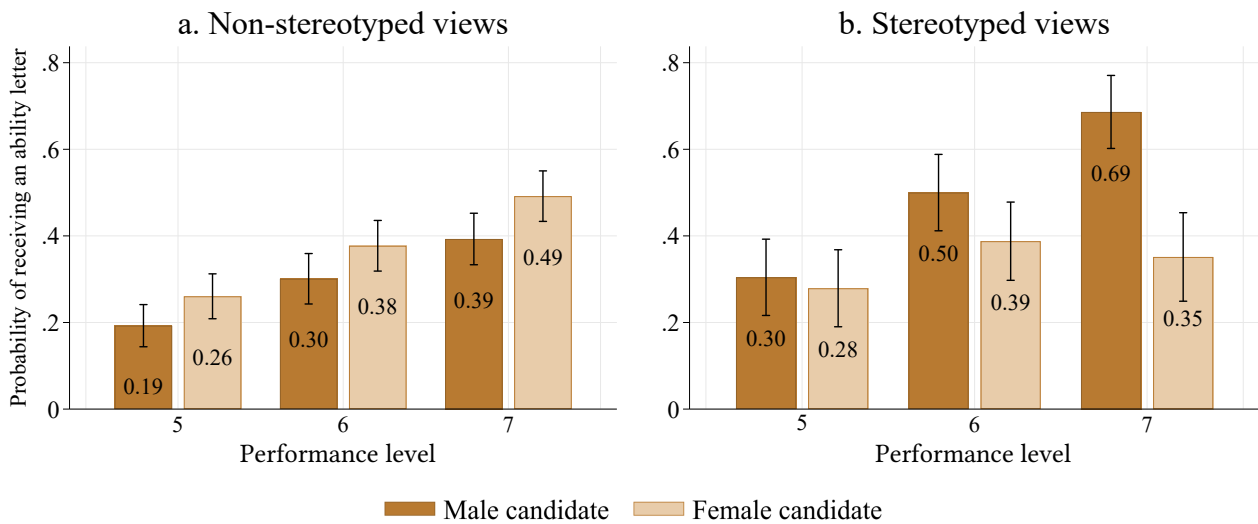
Even though previous research using observational data employs a rich set of candidate controls, it cannot completely rule out confounds about candidates’ quality. Our experimental approach, by contrast, directly addresses this concern. In the academic survey, candidates have identical observed characteristics by design. This makes male and female candidates highly comparable, weakening the argument that differences in letters could be driven by actual gender differences in ability or effort. However, a remaining concern is that academic recommenders might still draw different inferences about a male or female candidate’s potential from the identical observed characteristics. This might

happen as performance is not objectively measured, and as a result recommenders might draw on their experience to interpret the provided information.

In the online experiment, we go one step further and rule out such concerns. We provide recommenders with direct information on an objective performance measure. Nevertheless, we find that gender differences in letters persist in both environments, making it unlikely that actual differences in ability and hard work across genders play an important role. We also do not see any significant differences in quiz performance or cognitive skill between male and female candidates, nor differences in recommenders' or recruiters' expectations about their performance (see Appendix Figures A.3–A.5). The only dimension on which female and male candidates differ is average self-reported effort, with men reporting significantly higher levels (*ibid.*,  $z = 2.74$ ,  $p = 0.006$ ). If anything, this would justify more grindstone letters for male candidates. However, this result should be interpreted carefully as there may be gender differences in confidence or reporting (e.g., Niederle and Vesterlund, 2007; Adamecz-Völgyi and Shure, 2022).<sup>17</sup>

Finally, the online experiment allows us to assess whether recommenders react differently to candidate quality depending on gender views. Figure 9 shows the share of male and female candidates receiving an ability letter by performance level, separately for recommenders who hold (respectively, do not hold) stereotyped views. Among recommenders with non-stereotyped views (panel a.), there is a clear positive relationship between performance and the likelihood of receiving an ability letter. Among recommenders with stereotyped views (panel b.), by contrast, this relationship only holds for male candidates.

Figure 9: Probability of receiving an ability letter by performance level



Taken together, the analysis in this section sheds light on the role of candidate quality in explaining the type of recommendation candidates receive. It is not that the distribution of quality is different

<sup>17</sup>In fact, we find that the correlation between reported effort and time spent on the quizzes is slightly stronger for women ( $r = 0.5$  versus  $r = 0.46$ ).

between men and women; instead, recommenders who hold stereotyped views react differently to the same quality depending on the gender of the candidate.<sup>18</sup>

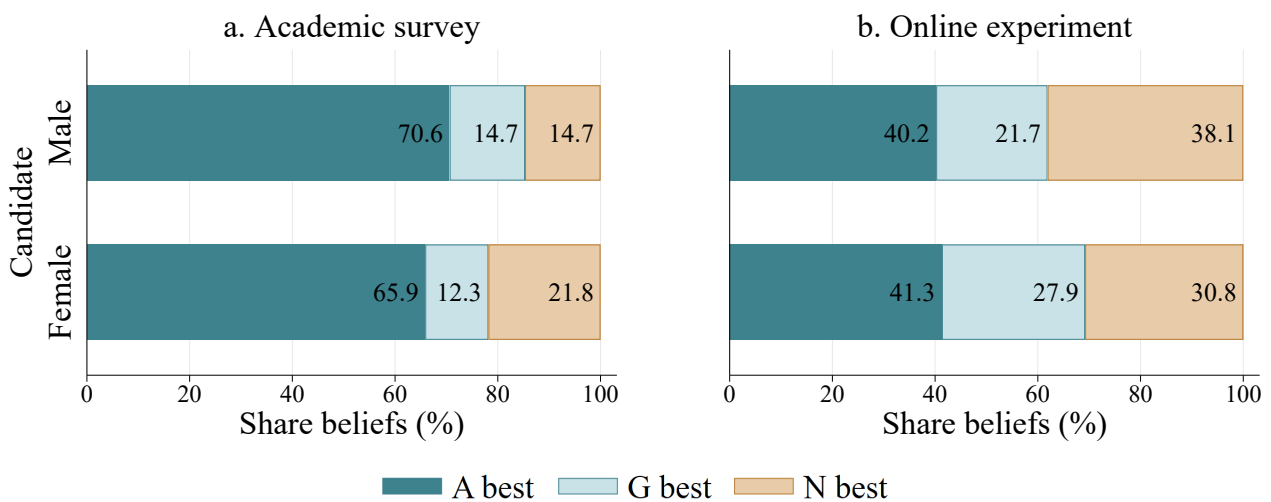
## 6.2 Strategic Considerations and Beliefs

A final reason why men and women receive different letters is that recommenders may believe that it is beneficial for women to be described as hard-working or to understate their abilities. Recommenders may for instance think that a message strongly emphasizing ability will portray women as unlikable (e.g., [Cooper, 2013](#)). To gauge the importance of this mechanism, we conduct two analyses.

Both the academic survey and the online experiment contain a ‘hiring’ component that allows us to test whether it is optimal for female and male candidates to receive different letters. We showed in [Section 4](#) that the majority of recruiters prefer to hire a candidate with an ability letter (see [Figure 4](#)). Moreover, [Figure 5](#) highlights that, if anything, this effect is more pronounced for female candidates. Similarly, receiving a grindstone message is slightly worse for them. These patterns do not support the view that catering to recruiters’ preferences can explain why men and women receive different letters.

Although it is not optimal for female candidates to receive fewer ability letters, recommenders may not know this. To study this possibility, we elicit recommenders’ beliefs about the most effective letter. We do so by asking them to predict which letter maximizes the candidate’s chance of being hired. As we just discussed, the correct answer to this question is the ability letter. However, [Figure 10](#) reveals that many recommenders do not choose the right answer. A non-negligible share thinks that most recruiters will prefer to hire a candidate with a grindstone or a neutral letter. Moreover, in the

Figure 10: Beliefs about the most effective letter



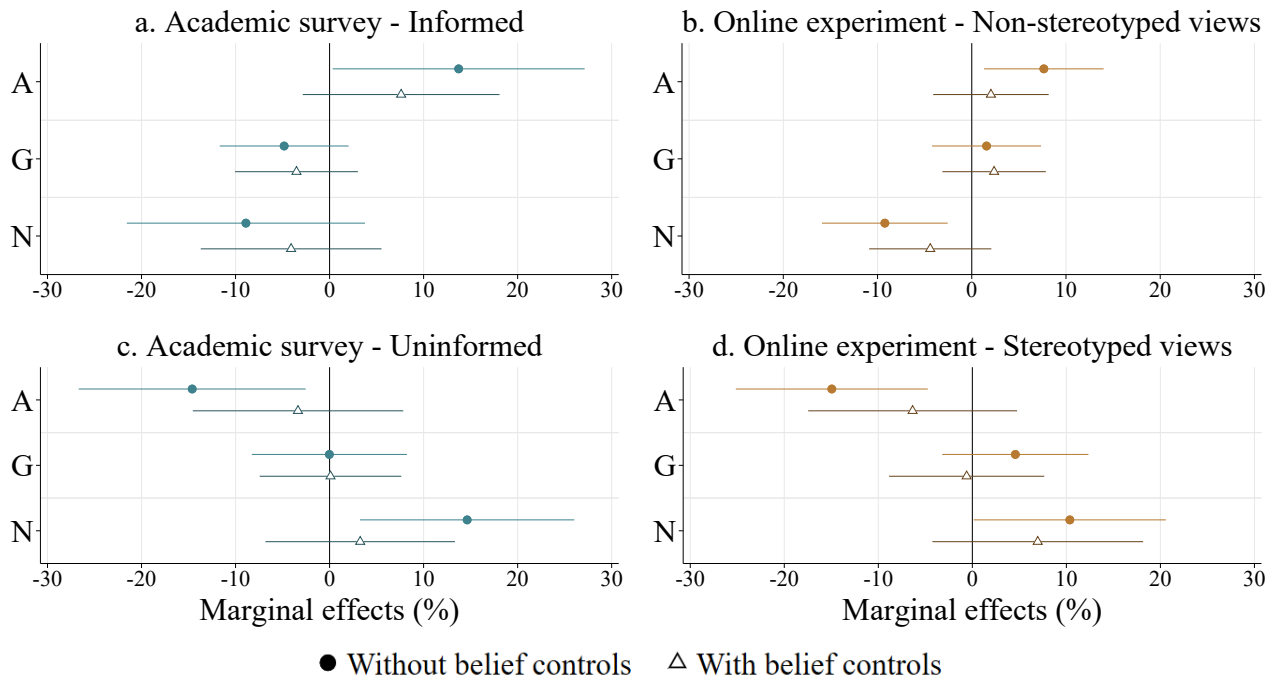
*Note:* Share of recommenders believing a given letter maximizing the candidate’s probability to be hired. In reality, the best letter for a candidate (independent of gender) is the ability one.

<sup>18</sup>Previous research shows that gendered reactions to individuals’ signals or behavior can continue over time, potentially generating cumulative effects (e.g., [Egan et al., 2022](#); [Sarsons, 2022](#)).

academic survey, this error is more pronounced for female candidates. Although an ability letter is *more* effective for women than men, Panel a. of Figure 10 reveals that *fewer* recommenders select the correct answer for women compared to men.

Given these inconsistencies between the recommenders' beliefs and the recruiters' hiring behavior, we explore whether accounting for beliefs can explain the gender gap in letter choice. To do so, we control for these beliefs in the letter choice regressions. In particular, we add binary control variables flagging which of the letters was selected as the one maximizing the candidate's chances of being recruited (see Appendix Tables A.12 and A.13). Figure 11 presents the results from our baseline models alongside those in which we control for recommenders' beliefs. Across all studies and sub-samples, adding beliefs as a control attenuates the size of the gender effect on letter choice, which is now always statistically insignificant. Moreover, we use nonparametric bootstrapping to test whether the difference between marginal effects in models with and without belief controls is statistically significant. We find that the marginal effects of gender differ significantly across model specifications with and without belief controls.<sup>19</sup> In line with Hypothesis 3, these results suggest that strategic considerations and erroneous beliefs are a key driver of letter choices.

Figure 11: Differences in letters between female and male candidates after controlling for beliefs



*Note:* The figure plots marginal effects from multinomial logit regressions, where the dependent variable is 0 if a neutral letter (N) was chosen, 1 if a grindstone letter (G) was chosen, and 2 if an ability letter (A) was chosen. Regressions control for order effects and include recommender controls. In each panel, the same regression is once run with and without controlling for recommenders' beliefs. Whiskers represent 90% confidence intervals. See Appendix tables A.12 and A.13 for the full set of estimates. Results are qualitatively the same when using a weighted online sample (see Appendix Table A.14).

<sup>19</sup>Among recommenders with stereotyped views, the effect is significant for ability letters ( $z = -2.41$ ,  $p = 0.016$ ). For recommenders with non-stereotyped views, significant differences are found for both ability ( $z = 3.36$ ,  $p = 0.001$ ) and neutral letters ( $z = -2.88$ ,  $p = 0.004$ ). Similarly, for recommenders with uninformed views, we observe significant differences for ability and neutral letters ( $z = -2.5$ ,  $p = 0.013$  and  $z = 2.46$ ,  $p = 0.014$ ).

A separate question is whether recommenders either act with the candidate’s best interest at heart or hold motivated beliefs (e.g., [Bénabou, 2015](#)). Although the majority of recommenders send recruiters what they believe to be the most effective letter, these beliefs also correlate with gender views (see Figure [A.6](#)). In particular, recommenders with uninformed/stereotyped views are more likely to hold the erroneous belief that ability letters are *better* for men than for women. This alignment between gender views and beliefs is broadly consistent with motivated reasoning, with recommenders developing a narrative that justifies their choices.

Independently of the source of beliefs, our results suggest that informing recommenders about the effectiveness of ability letters could be a promising approach to decreasing differences in letters written for female and male candidates. Recent evidence suggests this also applies in the case of motivated beliefs (e.g., [Zimmermann, 2020](#)).

## 7 Discrimination Decomposition

The discussion of gender representation in the workplace has been taken seriously by many companies and organizations, for instance through DEI policies. In the context of recommendations, we have documented systematic differences in treatment that may undermine these efforts. Our analysis has focused on two instances: the choice of letters on the one hand, and the hiring of candidates conditional on having received the same letter on the other.

An important next step is to understand the interplay of these two instances, and their implications for diversity in hiring. More specifically, despite existing policies to promote the recruitment of women, hiring decisions may still be distorted if reference letters differ systematically for men and women.

To conceptualize differential treatment at different points in the hiring process, we apply the classic Kitagawa-Oaxaca-Blinder decompositions ([Kitagawa 1955](#); [Oaxaca 1973](#); [Blinder 1973](#)) to break down ‘total discrimination’ into a ‘direct’ and ‘systemic’ component (see [Bohren et al. 2025](#), whose terminology we draw on in this section, and [Baron et al. 2024](#) for an application). In our context, favoring female over male applications with identical attributes is ‘direct discrimination’. In contrast, gender gaps caused by recruitment decisions blindly relying on reference letters without realizing that these differ for men and women is considered as ‘systemic discrimination’. Below, we outline the decomposition framework and how we apply it to our data.

### 7.1 Theoretical Framework

Denote  $j \in \mathcal{J}$  the recruiters, and  $i \in \mathcal{I}$  the candidates, whose gender  $G_i \in \{F, M\}$  is observed by recruiters (with F denoting female and M male). Candidates have ex-ante productivity  $Y_i$ . Recruiters receive a signal of this productivity through the reference letters  $L_i \in \{\text{Ability, Grindstone, Neutral}\}$  and decide on a recruitment outcome  $\text{Recruit}_j(G_i, L_i) \in \{0, 1\}$ .

### 7.1.1 Total Discrimination

Total Discrimination (TD) is the expected difference in recruitment outcomes between male and female candidates, conditional on productivity  $y$ . This is the empirical hiring gap observed by the researcher and can be written as:

$$TD(y) = \mathbb{E}[\text{Recruit}_j(G_i, L_i)|G_i = F, Y_i = y] - \mathbb{E}[\text{Recruit}_j(G_i, L_i)|G_i = M, Y_i = y]. \quad (1)$$

### 7.1.2 Direct Discrimination

Direct discrimination is the difference in recruitment outcomes between male and female candidates with identical letters. In other words, direct discrimination arises when recruiters do not apply the same hiring rule for men and women upon receiving the same letter. Formally, a measure of direct discrimination by recruiter  $j$  after receiving a given letter  $l$  is as follows:

$$\tau_j(l) = \text{Recruit}_j(F|L_i = l) - \text{Recruit}_j(M|L_i = l). \quad (2)$$

The Average Direct Discrimination (ADD for short) is the average of  $\tau(l)$  over the letter distribution in the population.<sup>20</sup> Formally,

$$ADD(g, y) = \mathbb{E}[\tau(L_i)|G_i = g, Y_i = y]. \quad (3)$$

### 7.1.3 Systemic Discrimination

Finally, Systemic Discrimination (SD) captures the disparities in recruitment outcomes because recruiters ignore that letter types are correlated with candidate gender. A measure of systemic discrimination at productivity  $y$  is given by:

$$SD(g, y) = \mathbb{E}[\text{Recruit}_j(g, L_i)|G_i = F, Y_i = y] - \mathbb{E}[\text{Recruit}_j(g, L_i)|G_i = M, Y_i = y]. \quad (4)$$

When  $g = M$ , the second term on the right-hand side of the equation gives the expected realized recruitment outcomes for men (as in the second term on the right-hand side of equation 1). The first term gives the expected recruitment outcomes for women if they are assessed using the recruitment rules for men (i.e., shutting down any channel of direct discrimination based on gender, but not taking into account that the distribution of letters differs for men and women). The two terms can only differ if the underlying distribution of letter types is not the same across genders. In other words,  $SD(M, y)$  captures recruitment gaps that arise only as a result of differences in the distribution of letters.

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<sup>20</sup>Alternatively, we can calculate ADD using the letter distribution just for the female, or just the male population. All three specifications produce qualitatively similar results. We refer to the respective tables in the appendix when discussing our results.



### 7.1.4 Decomposition

Combining equations 1 and 4, we can decompose total discrimination as the sum of the average direct discrimination and systemic discrimination. Specifically,

$$\begin{aligned}
TD(y) &= \mathbb{E}[\text{Recruit}_j(F, L_i)|G_i = F, Y_i = y] - \mathbb{E}[\text{Recruit}_j(M, L_i)|G_i = M, Y_i = y] \\
&\quad + \mathbb{E}[\text{Recruit}_j(M, L_i)|G_i = F, Y_i = y] - \mathbb{E}[\text{Recruit}_j(M, L_i)|G_i = F, Y_i = y] \\
&= \underbrace{\mathbb{E}[\text{Recruit}_j(F, L_i) - \text{Recruit}_j(M, L_i)|G_i = F, Y_i = y]}_{\text{Average Direct Discrimination for Women } ADD(F, y)} \\
&\quad + \underbrace{\mathbb{E}[\text{Recruit}_j(M, L_i)|G_i = F, Y_i = y] - \mathbb{E}[\text{Recruit}_j(M, L_i)|G_i = M, Y_i = y]}_{\text{Systemic Discrimination } SD(M, y)} \\
&= ADD(F, y) + SD(M, y).
\end{aligned} \tag{5}$$

## 7.2 Estimation and Results

We now turn to the estimation of total, direct, and systemic discrimination. In addition, we evaluate heterogeneity with respect to gender views. Given the importance of strategic considerations and beliefs discussed in Section 5, we also analyze a hypothetical scenario that assumes all recommenders to have correct/incorrect beliefs.

To obtain these estimates, we first need to link recommenders and recruiters. In both the academic survey and the online experiment, recruiters see an equal number of ability, grindstone, and neutral letters. However, to understand the total effect, we need to account for the distribution of letters generated by recommenders. To do so, we run 1,000 simulations per specification, randomly matching one recommender and their letter choice to one recruiter. We determine whether each match between a recruiter and a letter would result in a successful hire, based on the recruiter's decision in the experiment.

### 7.2.1 Estimation

**Total Discrimination.** TD is the expected difference in recruitment outcomes between male and female candidates. Since gender is randomized in our two experiments, we can estimate this difference by regressing the hiring outcome on gender:

$$\text{Recruit}_{il} = \alpha + \beta \text{Female}_i + \varepsilon_{il}, \tag{6}$$

where  $\text{Recruit}_{il}$  is a binary variable which takes the value of 1 if a candidate  $i$  with letter  $l$  is hired and 0 otherwise;  $\text{Female}_i$  is a binary variable equal to one if the candidate is a woman. The coefficient  $\hat{\beta}$  provides an estimate for TD.

**Average Direct Discrimination.** ADD captures the average gender difference in recruitment outcomes for candidates with the same letter. To quantify ADD we estimate the following regression

equation using Ordinary Least Squares (OLS):

$$\text{Recruit}_{id} = \sum_{k=1}^3 (\gamma_k \mathbf{1}\{L_i = l_k\} + \lambda_k \mathbf{1}\{L_i = l_k\} \times \text{Female}_i) + \nu_{id}, \quad (7)$$

where  $\mathbf{1}\{L_i = l_k\}$  is a binary variable equal to one if the letter for candidate  $i$  is of type  $l_k \in \{\text{Ability, Grindstone, Neutral}\}$ .<sup>21</sup>

If the classical OLS assumptions hold, then  $\lambda_k$  gives us the average direct discrimination *for letter type*  $l_k$  ( $\mathbb{E}[\tau(l_k)]$ ) (see equation 2, where  $k$  was omitted for simplicity of notation). An estimator of the average direct discrimination is then the average of  $\hat{\lambda}_k$ , weighted by the probability of each letter type:

$$\widehat{\text{ADD}}(g) = \sum_{k=1}^3 \hat{\lambda}_k \hat{p}(l_k | \text{Female}_i \in g). \quad (8)$$

We estimate the probability of each letter type  $p(l_k | \text{Female}_i = g)$  using the distribution of letter types in the sample of all candidates.

**Systemic Discrimination.** Systemic Discrimination is the difference between Total and Average Direct Discrimination ( $\hat{\beta} - \widehat{\text{ADD}}(g)$ ).

Before presenting the results of our decomposition exercise, it is worth noting two specific features of our experimental studies.

First, in the academic survey, we ask recruiters which letter they find most convincing, rather than which candidate they would hire. While this does not rule out the possibility that recruiters hire candidates with weaker letters, given the competitiveness of the academic job market, we regard recruiters' relative preferences as a meaningful proxy for actual hiring decisions. Moreover, this concern does not apply to the online experiment, where recruiters make direct hiring choices among candidates with different letters.

Second, we do not elicit hiring decisions for both male and female candidates from the same recruiter. Although this between-subject design may introduce some noise due to individual variability, crucially, it minimizes experimenter demand effects while enabling us to detect average treatment effects, which we decompose in the following section.

### 7.2.2 Results

Table 1 shows the results of the decomposition exercise. Columns 1 and 4 give results for the full sample in the academic survey and online experiment, respectively, while columns 2-3 and 5-6 con-

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<sup>21</sup>The regression has no constant because the sum of all  $\mathbf{1}\{L_i = l_k\}$  equals the constant. The regression does not include the female binary variable because the sum of all  $\mathbf{1}\{L_i = l_k\} \times \text{Female}_i$  equals the female binary.

Table 1: Decomposition of total into systemic and direct discrimination - Gender views

|              | <i>Academic survey</i> |                              |           | <i>Online experiment</i> |                                |            |
|--------------|------------------------|------------------------------|-----------|--------------------------|--------------------------------|------------|
|              | (1)<br>Full sample     | (2)<br>Informed views<br>Yes | (3)<br>No | (4)<br>Full sample       | (5)<br>Stereotyped views<br>No | (6)<br>Yes |
| TD           | 1.55                   | 8.08                         | -4.70     | 1.34                     | 4.13                           | -5.21      |
| ADD          | 3.52                   | 3.86                         | 2.95      | 0.38                     | 0.10                           | 0.99       |
| SD           | -1.97                  | 4.22                         | -7.65     | 0.96                     | 4.03                           | -6.19      |
| Recommenders | 506                    | 232                          | 274       | 959                      | 679                            | 280        |
| Recruiters   | 514                    | 514                          | 514       | 959                      | 959                            | 959        |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column. It presents results using the distribution of letters for all candidates. Tables A.15 and A.16 show that results are qualitatively and in terms of magnitude identical when using the distribution for only female or male candidates or using weighted regressions for the online sample. TD, ADD and SD are Total, Average Direct and Systemic Discrimination, respectively.

trast results depending on gender views.

The results show that for the full sample in the academic survey, TD is small but *positive*: overall, female candidates have a 1.55 p.p. *higher* chance of being hired compared to their male counterparts. ADD in favor of women is more than twice as large (3.52). However, this is counter-acted by a *negative* SD stemming from women receiving worse letters on average. When focusing on the sub-sample of recommenders who have uninformed views (column 3), the negative SD is exacerbated and becomes even larger than the positive ADD. If recruiters only receive letters from recommenders who have uninformed views, female candidates face an overall 4.70 p.p. lower probability of being hired. By contrast, if recommenders have informed views (column 2), SD becomes positive, resulting in a TD in favor of women of 8.08 p.p.

The online experiment shows the same qualitative patterns, with positive ADD in favor of women throughout, while SD disadvantages women in the sub-sample of recommenders with stereotyped views and advantages them in the sub-sample with non-stereotyped views.<sup>22</sup>

In Table 2, we conduct the decomposition for recommenders with correct and incorrect beliefs, i.e. whether a recommender believes that an ability letter is best or not. Columns 1 and 4 report the full sample results, whereas columns 2-3 and 5-6 contrast findings depending on recommender beliefs. In Section 6, we show that the beliefs of recommenders affect the distribution of letters for men and women. Different distributions will in turn affect SD. In line with this argument, Table 2 shows that in the academic survey the gap in SD between recommenders holding correct and incorrect beliefs is large (columns 2 and 3: 3.36 vs. -7.98). This gap in turn flips the sign of TD (8.40 vs. -7.92). In the online experiment, a similar pattern emerges even though TD remains positive even under incorrect beliefs.

<sup>22</sup>Table A.17 in Appendix A.7 shows additional decomposition results when distinguishing between *recruiters* with different gender views. Results are qualitatively similar to using the full sample of recruiters.

Table 2: Decomposition of total into systemic and direct discrimination - Beliefs

|              | <i>Academic survey</i> |                               |           | <i>Online experiment</i> |                               |           |
|--------------|------------------------|-------------------------------|-----------|--------------------------|-------------------------------|-----------|
|              | (1)<br>Full sample     | (2)<br>Correct beliefs<br>Yes | (3)<br>No | (4)<br>Full sample       | (5)<br>Correct beliefs<br>Yes | (6)<br>No |
| TD           | 1.55                   | 8.40                          | -7.92     | 1.34                     | 2.35                          | 0.36      |
| ADD          | 3.52                   | 5.05                          | 0.07      | 0.38                     | 2.11                          | -0.83     |
| SD           | -1.97                  | 3.36                          | -7.98     | 0.96                     | 0.24                          | 1.20      |
| Recommenders | 506                    | 344                           | 160       | 959                      | 391                           | 568       |
| Recruiters   | 514                    | 514                           | 514       | 959                      | 959                           | 959       |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column. Correct beliefs restricts the analysis to recommenders who correctly believe that the ability letter will be most successful. In the academic survey, two recommenders did not answer the beliefs question. The table presents results using the distribution of letters for all candidates. Tables A.18 and A.19 show that results are qualitatively and in terms of magnitude identical when using the distribution for only female or male candidates or using weighted regressions for the online sample. TD, ADD and SD are Total, Average Direct and Systemic Discrimination, respectively.

Overall, these findings highlight the importance of taking systemic elements into account when assessing discrimination patterns. The positive ADD we find in all specifications is indicative of the widespread efforts to improve female recruitment, in line with current DEI policies. However, when looking at final outcomes for female candidates (TD), these efforts may be insufficient if there are systemic elements that negatively affect women’s chances of being hired. In the full sample, the attempt to hire more women is almost counter-balanced by the negative SD. In the sample of recommenders having uninformed/stereotyped views, we see that SD is substantial enough to lead to an overall negative gender hiring gap (TD). Similarly, incorrect beliefs by recommenders undermine efforts to diversify hiring. The resulting SD can be large enough to drive a negative TD.

## 8 Concluding Remarks

In this paper, we study the gendered impact of recommendation content on recruitment, focusing on reference letters. Previous research has documented that men and women are described differently in reference letters on the academic job market for economists: female candidates are less often described in terms of ability, and more often characterized with so-called ‘grindstone’ attributes. We go beyond these empirical findings, aiming to understand the mechanisms underlying this difference and the consequences for hiring outcomes.

Using novel experimental data, we first show that, at the recruitment stage, women have, *ceteris paribus*, *higher* recruitment prospects compared to men, which is in line with recent DEI efforts. At the same time, independent of the candidates’ gender, recommendations that highlight ‘ability’ attributes increase the chances of being hired. This causal effect of recommendations on hiring outcomes is confirmed both in a survey experiment with academic economists as well as an online

experiment with a more general population. However, at the recommendation stage, we observe that some recommenders are *less* likely to choose ‘ability’ letters for women, undermining the diversification efforts at the hiring stage.

Our results highlight the role of strategic but erroneous beliefs and gender views as drivers of letter choices in the experiment. Recommenders aim to maximize the chances of a candidate to get hired, but underestimate the importance of ability letters. Similarly, recommenders with uninformed or stereotyped views are less likely to choose the best letter for women – a pattern that we further corroborate in a two-sample prediction exercise based on *JM* data, where predicted gender views similarly shape the choice of letter type.

Our experimental analysis, as well as results from Kitagawa-Oaxaca-Blinder decompositions, suggest that differences in recommendations received by men and women are a more significant barrier to women’s recruitment than outright unequal treatment of candidates with identical characteristics. In other words, to tackle barriers to diversity it is crucial to understand more subtle and indirect forms of discrimination, focusing not only on the recruiter side but also the recommender side of the market. This matters because a significant portion of the efforts to improve diversity in recruitment targets the hiring process, for example through training programs, quotas in interviewing panels, shortlisting targets, or systematic compiling and analyzing of diversity data for Human Resource departments. While recruiters could be trained to take gender imbalances in recommendations into account, a more direct approach would be to target recommenders. In our setting, the weak link in overcoming gender recruitment gaps is represented in this earlier step in the recruitment process, namely by recommenders emphasizing different qualities in their recommendations for male and female candidates. Therefore, correcting recommender beliefs about the effectiveness of different letter types or making them aware of their own – potentially uninformed or stereotyped – gender views could meaningfully improve diversity in the workplace.

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# Appendix

## Contents

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| <b>A</b> | <b>Additional analysis</b>                                            | <b>38</b> |
| A.1      | Descriptive Statistics . . . . .                                      | 38        |
| A.2      | Hiring Decisions . . . . .                                            | 40        |
| A.3      | Letter Choices . . . . .                                              | 42        |
| A.4      | Comparison to Economics Job Market data . . . . .                     | 44        |
| A.4.1    | Text analysis - Economic Job Market ( <i>JM</i> ) data . . . . .      | 44        |
| A.4.2    | Descriptives . . . . .                                                | 44        |
| A.4.3    | Analysis . . . . .                                                    | 46        |
| A.5      | Are male and female candidates different? . . . . .                   | 48        |
| A.6      | Strategic Considerations and Beliefs . . . . .                        | 49        |
| A.7      | Discrimination Decomposition . . . . .                                | 52        |
| <b>B</b> | <b>Instructions &amp; Procedures</b>                                  | <b>58</b> |
| B.1      | Sample construction academic survey . . . . .                         | 58        |
| B.2      | Instructions Academic survey . . . . .                                | 58        |
| B.2.1    | Consent . . . . .                                                     | 58        |
| B.2.2    | Recommender survey . . . . .                                          | 59        |
| B.2.3    | Recruiter Survey . . . . .                                            | 64        |
| B.2.4    | Questionnaire (common to recommender and recruiter surveys) . . . . . | 68        |
| B.3      | Full reference letter . . . . .                                       | 70        |
| B.4      | Instructions Online experiment . . . . .                              | 71        |
| B.4.1    | Workers . . . . .                                                     | 71        |
| B.4.2    | Recommenders . . . . .                                                | 79        |
| B.4.3    | Recruiters . . . . .                                                  | 86        |
| B.4.4    | Questionnaire for recommenders and recruiters . . . . .               | 92        |

## A Additional analysis

### A.1 Descriptive Statistics

Tables A.1 and A.2 give an overview of the demographic characteristics in the academic survey and the online experiment. In addition to the characteristics presented in Figure 3 in the main text, the tables show that 87% of participants have experience hiring on the junior academic job market and 77% as recommenders. In the online experiment, we required participants to have at least an

Table A.1: Summary Statistics — Academic Survey

|                                   | Mean     | SD     | N     |
|-----------------------------------|----------|--------|-------|
| Female                            | 0.26     | 0.44   | 1,018 |
| <i>Academic rank:</i>             |          |        |       |
| Assistant Professor               | 0.34     | 0.47   | 1,019 |
| Associate Professor               | 0.24     | 0.43   | 1,019 |
| Full Professor                    | 0.41     | 0.49   | 1,019 |
| RePEc rank                        | 291.89   | 206.62 | 1,019 |
| Year of PhD                       | 2,006.02 | 11.34  | 972   |
| RePEc Top100                      | 0.21     | 0.41   | 1,020 |
| <i>Hiring experience:</i>         |          |        |       |
| Never                             | 0.13     | 0.33   | 1,019 |
| 1-5                               | 0.50     | 0.50   | 1,019 |
| 6-10                              | 0.19     | 0.39   | 1,019 |
| 11-20                             | 0.12     | 0.32   | 1,019 |
| >20                               | 0.07     | 0.25   | 1,019 |
| <i>Letter writing experience:</i> |          |        |       |
| None                              | 0.23     | 0.42   | 1,017 |
| 1-5                               | 0.30     | 0.46   | 1,017 |
| 6-10                              | 0.18     | 0.38   | 1,017 |
| 11-20                             | 0.12     | 0.32   | 1,017 |
| _cons                             | 0.17     | 0.37   | 1,017 |
| Literature known                  | 0.48     | 0.50   | 1,020 |

Table A.2: Summary Statistics — online experiment

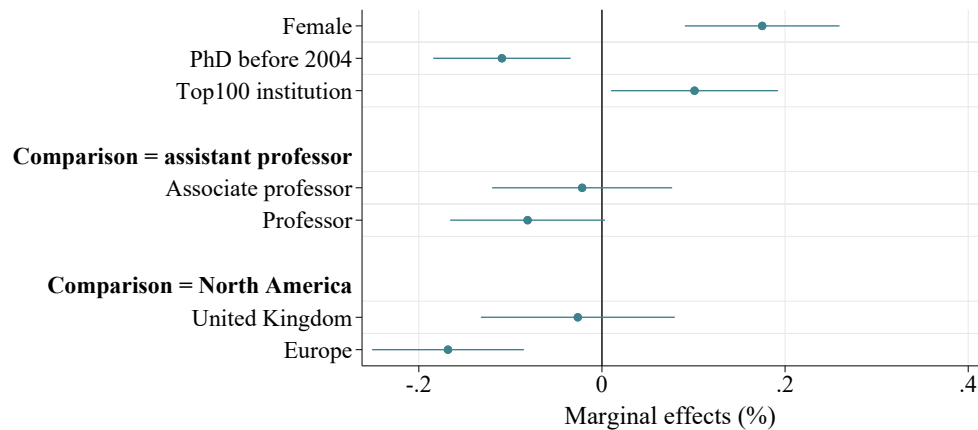
|                     | Mean  | SD    | N     |
|---------------------|-------|-------|-------|
| Female              | 0.50  | 0.50  | 1,913 |
| Age                 | 39.20 | 12.88 | 1,918 |
| <i>Education:</i>   |       |       |       |
| High school         | 0.01  | 0.08  | 1,918 |
| Undergraduate       | 0.67  | 0.47  | 1,918 |
| Masters             | 0.27  | 0.45  | 1,918 |
| PhD                 | 0.04  | 0.21  | 1,918 |
| <i>Nationality:</i> |       |       |       |
| UK                  | 0.89  | 0.31  | 1,711 |
| US                  | 0.07  | 0.25  | 1,711 |
| Others              | 0.04  | 0.19  | 1,711 |
| Stereotyped views   | 0.28  | 0.45  | 1,918 |



undergraduate degree, which is also the most common educational degree in the sample (67%).

In the *academic survey*, we distinguish between participants with informed and uninformed views based on their knowledge of recent literature on gender differences in hiring. Figure A.1 shows which demographic characteristics in the academic survey correlate with participants having uninformed views.

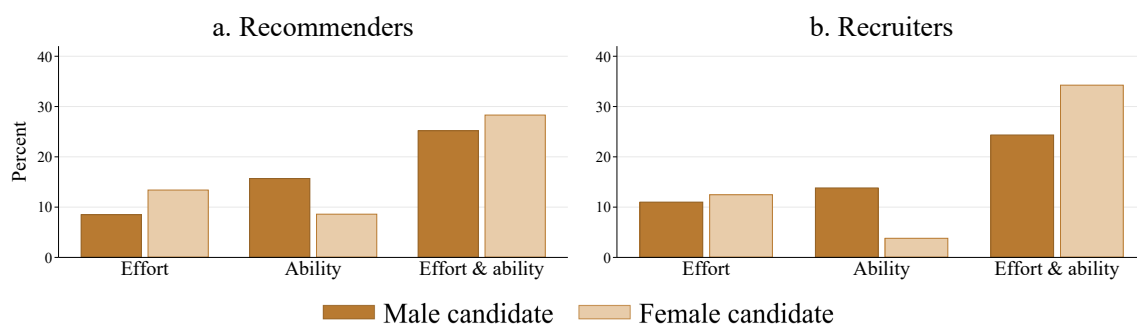
Figure A.1: Correlates of gender views in the academic survey



*Note:* The figure shows marginal effects from regressing gender views on demographic characteristics in the academic survey. Each estimate comes from a separate regression using only one explanatory variable. The dependent variable is 0 if the participant is classified as having uninformed views and 1 if they are classified as having informed views. Whiskers represent 90% confidence intervals.

In the *online experiment*, we distinguish between participants with stereotyped and non-stereotyped views. To do so, we ask recruiters whether male/female workers are characterized more by ability, effort, or equally possess both traits. Recommenders are asked to indicate what opinion they think most recruiters have. Figure A.2 shows how participants answered when asked about male or female candidates. We see a similar pattern for both recommenders and recruiters. Male candidates are much more likely to be perceived as mainly characterized by ability, while female candidates are more likely to be mainly seen as characterized by effort. Participants who express these opinions are defined as having stereotyped views in our study.

Figure A.2: Perception of men and women in the online experiment



*Note:* We asked participants whether they thought that women/men tend to be characterized more by high effort, high ability or equally by high effort and high ability.

## A.2 Hiring Decisions

Columns 1 and 2 in Table A.3 present results from a multinomial logit regression of hiring decisions on candidate gender and other control variables in the academic survey. Table A.4 does the same for the online experiment. These results are presented in Figure 5 of the main text.

In addition to looking at the full sample, we explore whether hiring decisions depend on the recruiters' gender views. Columns 3 and 4 show results for the sub-sample of informed/non-stereotyped recruiters, while columns 5 and 6 show results for uninformed/stereotyped recruiters. We see that the same qualitative pattern holds in both groups, but some differences emerge. In the online experiment, the effects are larger in magnitude for participants with stereotyped views. In the academic survey, recruiters who have informed gender views place a larger penalty on grindstone letters for women relative to men, while those who have uninformed views place a larger premium on ability letters for women relative to men.

In the main text, we do not weight the experimental sample. As men are often more likely to be in positions of power, our results are thus a more conservative estimate for any gender differences. As a robustness check, we re-weight the experimental sample using the gender distribution in the observational sample as the underlying distribution. Our findings on the hiring side are presented in Table A.5. Results indeed become stronger/more differentiated after re-weighting. We find that women are now significantly less likely to be hired with a grindstone or neutral letter compared to men in the full sample.

Table A.3: Hiring choice in the academic survey - Weighted multinomial logits

|                               | <b>Full sample</b> |                   | <b>Informed</b>   |                   | <b>Uninformed</b> |                  |
|-------------------------------|--------------------|-------------------|-------------------|-------------------|-------------------|------------------|
|                               | (1)                | (2)               | (3)               | (4)               | (5)               | (6)              |
| Ability - Female candidate    | 6.98<br>(5.62)     | 8.59<br>(5.53)    | 4.61<br>(8.48)    | 6.87<br>(8.45)    | 10.21<br>(7.07)   | 11.78*<br>(6.20) |
| Grindstone - Female candidate | -7.88*<br>(4.67)   | -9.39**<br>(4.63) | -12.54*<br>(7.32) | -14.32*<br>(7.31) | -3.92<br>(5.61)   | -5.79<br>(4.75)  |
| Neutral - Female candidate    | 0.90<br>(4.05)     | 0.80<br>(4.03)    | 7.93<br>(5.24)    | 7.45<br>(5.52)    | -6.28<br>(5.83)   | -5.99<br>(5.24)  |
| Observations                  | 513                | 513               | 257               | 257               | 256               | 256              |
| Referrer controls             | No                 | Yes               | No                | Yes               | No                | Yes              |
| Pseudo R <sup>2</sup>         | 0.10               | 0.12              | 0.07              | 0.11              | 0.16              | 0.21             |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a candidate with a neutral letter was chosen, 1 if a candidate with a grindstone letter was chosen, and 2 if a candidate with an ability letter was chosen. Recruiter controls include gender, academic position, institutional rank, letter writing experience, and hiring experience. All regressions control for order effects.

Table A.4: Hiring choice in the online experiment - multinomial logits

|                               | <b>Full sample</b> |                 | <b>Non-stereotyped</b> |                  | <b>Stereotyped</b> |                   |
|-------------------------------|--------------------|-----------------|------------------------|------------------|--------------------|-------------------|
|                               | (1)                | (2)             | (3)                    | (4)              | (5)                | (6)               |
| Ability - Female candidate    | 6.46**<br>(3.16)   | 6.09*<br>(3.15) | 6.54*<br>(3.66)        | 5.87<br>(3.64)   | 7.83<br>(6.38)     | 8.88<br>(6.42)    |
| Grindstone - Female candidate | -3.63<br>(2.94)    | -3.27<br>(2.92) | -1.92<br>(3.36)        | -1.36<br>(3.35)  | -9.54<br>(6.04)    | -11.19*<br>(5.92) |
| Neutral - Female candidate    | -2.84<br>(2.15)    | -2.83<br>(2.19) | -4.61*<br>(2.48)       | -4.51*<br>(2.51) | 1.71<br>(4.54)     | 2.32<br>(4.68)    |
| Observations                  | 959                | 942             | 706                    | 697              | 253                | 245               |
| Recruiter controls            | No                 | Yes             | No                     | Yes              | No                 | Yes               |
| Pseudo R <sup>2</sup>         | 0.01               | 0.04            | 0.01                   | 0.05             | 0.03               | 0.09              |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a candidate with a neutral letter was chosen, 1 if a candidate with a grindstone letter was chosen, and 2 if a candidate with an ability letter was chosen. Recruiter controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. All regressions control for order effects.

Table A.5: Hiring choice in the online experiment - weighted multinomial logits

|                               | <b>Full sample</b> |                    | <b>Non-stereotyped</b> |                    | <b>Stereotyped</b> |                   |
|-------------------------------|--------------------|--------------------|------------------------|--------------------|--------------------|-------------------|
|                               | (1)                | (2)                | (3)                    | (4)                | (5)                | (6)               |
| Ability - Female candidate    | 11.91***<br>(3.69) | 11.79***<br>(3.71) | 11.93***<br>(4.19)     | 11.44***<br>(4.25) | 12.72<br>(7.82)    | 15.85**<br>(7.86) |
| Grindstone - Female candidate | -6.34*<br>(3.46)   | -6.02*<br>(3.46)   | -5.87<br>(3.90)        | -5.18<br>(3.92)    | -7.37<br>(7.45)    | -10.24<br>(7.41)  |
| Neutral - Female candidate    | -5.57**<br>(2.51)  | -5.77**<br>(2.51)  | -6.07**<br>(2.79)      | -6.26**<br>(2.82)  | -5.35<br>(5.60)    | -5.62<br>(5.73)   |
| Observations                  | 959                | 942                | 706                    | 697                | 253                | 245               |
| Recruiter controls            | No                 | Yes                | No                     | Yes                | No                 | Yes               |
| Pseudo R <sup>2</sup>         | 0.02               | 0.05               | 0.02                   | 0.06               | 0.05               | 0.10              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a candidate with a neutral letter was chosen, 1 if a candidate with a grindstone letter was chosen, and 2 if a candidate with an ability letter was chosen. Recruiter controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. Sampling weights are calculated taking the observational sample as the underlying population. All regressions control for order effects.

### A.3 Letter Choices

Table A.6 presents results from a multinomial logit regression of letter choices on candidate gender and other control variables in the academic survey. Table A.7 does the same for the online experiment. These results are illustrated in Figure 7 of the main text.

Table A.6: Letter choice in academic survey - Weighted multinomial logits

|                               | <b>Full sample</b> |                 | <b>Informed</b> |                  | <b>Uninformed</b>  |                    |
|-------------------------------|--------------------|-----------------|-----------------|------------------|--------------------|--------------------|
|                               | (1)                | (2)             | (3)             | (4)              | (5)                | (6)                |
| Ability - Female candidate    | -4.73<br>(6.06)    | -3.05<br>(6.04) | 10.04<br>(8.56) | 13.73*<br>(8.15) | -15.59**<br>(7.45) | -14.61**<br>(7.34) |
| Grindstone - Female candidate | -0.98<br>(3.69)    | -2.05<br>(3.65) | -2.76<br>(5.39) | -4.83<br>(4.17)  | -0.82<br>(4.95)    | -0.02<br>(5.01)    |
| Neutral - Female candidate    | 5.72<br>(5.71)     | 5.10<br>(5.65)  | -7.28<br>(8.15) | -8.90<br>(7.70)  | 16.41**<br>(7.07)  | 14.63**<br>(6.93)  |
| Observations                  | 505                | 505             | 231             | 231              | 274                | 274                |
| Referrer controls             | No                 | Yes             | No              | Yes              | No                 | Yes                |
| Pseudo R <sup>2</sup>         | 0.03               | 0.04            | 0.05            | 0.12             | 0.06               | 0.08               |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Recommender controls include gender, academic position, institutional rank, letter writing experience, and hiring experience. All regressions control for order effects.

Table A.8 uses the same specification as Table A.7 but employs weights based on the gender distribution in the observational sample. Again, marginal effects are larger and more precisely estimated

Table A.7: Letter choice in the online experiment - multinomial logits

|                               | <b>Full sample</b> |                 | <b>Non-stereotyped</b> |                   | <b>Stereotyped</b>  |                    |
|-------------------------------|--------------------|-----------------|------------------------|-------------------|---------------------|--------------------|
|                               | (1)                | (2)             | (3)                    | (4)               | (5)                 | (6)                |
| Ability - Female candidate    | 0.68<br>(3.06)     | 0.41<br>(3.37)  | 8.25**<br>(3.53)       | 7.65**<br>(3.86)  | -15.62***<br>(5.51) | -14.94**<br>(6.21) |
| Grindstone - Female candidate | 3.77<br>(2.64)     | 2.84<br>(2.85)  | 2.01<br>(3.23)         | 1.57<br>(3.52)    | 6.93<br>(4.28)      | 4.57<br>(4.73)     |
| Neutral - Female candidate    | -4.45<br>(3.15)    | -3.25<br>(3.49) | -10.25***<br>(3.68)    | -9.22**<br>(4.05) | 8.70<br>(5.66)      | 10.37*<br>(6.22)   |
| Observations                  | 959                | 769             | 679                    | 555               | 280                 | 214                |
| Referrer controls             | No                 | Yes             | No                     | Yes               | No                  | Yes                |
| Pseudo R <sup>2</sup>         | 0.03               | 0.06            | 0.03                   | 0.06              | 0.06                | 0.14               |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Recommender controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. All regressions control for order effects and the candidate's performance in Quiz A.

when using weights. In the sub-sample of participants with stereotyped views, women are now not only significantly less likely to receive an ability letter but also significantly more likely to receive a grindstone or neutral one.

Table A.8: Letter choice in the online experiment - weighted multinomial logits

|                               | <b>Full sample</b> |                 | <b>Non-stereotyped</b> |                    | <b>Stereotyped</b>  |                     |
|-------------------------------|--------------------|-----------------|------------------------|--------------------|---------------------|---------------------|
|                               | (1)                | (2)             | (3)                    | (4)                | (5)                 | (6)                 |
| Ability - Female candidate    | 1.72<br>(3.87)     | 0.22<br>(4.26)  | 10.31**<br>(4.40)      | 8.42*<br>(4.76)    | -21.56***<br>(6.76) | -26.52***<br>(6.89) |
| Grindstone - Female candidate | 3.24<br>(3.07)     | 2.67<br>(3.24)  | 1.00<br>(3.72)         | 1.76<br>(3.91)     | 10.20**<br>(4.78)   | 6.68<br>(4.46)      |
| Neutral - Female candidate    | -4.96<br>(3.86)    | -2.89<br>(4.27) | -11.31**<br>(4.44)     | -10.18**<br>(4.80) | 11.35<br>(7.02)     | 19.84***<br>(7.21)  |
| Observations                  | 954                | 769             | 675                    | 555                | 279                 | 214                 |
| Referrer controls             | No                 | Yes             | No                     | Yes                | No                  | Yes                 |
| Pseudo R <sup>2</sup>         | 0.02               | 0.06            | 0.03                   | 0.07               | 0.07                | 0.16                |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Recommender controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. Sampling weights are calculated by taking the observational sample as the underlying population. All regressions control for order effects and the candidate's performance in Quiz A.

## A.4 Comparison to Economics Job Market data

### A.4.1 Text analysis - Economic Job Market (JM) data

We take advantage of a unique collection of reference letters assembled by [Eberhardt et al. \(2023\)](#) on the basis of applications for assistant professor positions received by a UK research-intensive institution during the 2017-2021 period.<sup>23</sup> This collection spans almost 9,000 reference letters, written by over 4,000 recommenders in support of 2,900 candidates.<sup>24</sup>

The reference letters are transformed into quantitative data using the *term frequency inverse document frequency* (tf-idf) for each word, which quantifies the importance of a term in each letter compared to its prevalence in all letters.<sup>25</sup> The focus is on the final paragraph of each reference letter because it summarizes the candidate's skills, strengths, and prospects. To capture the 'sentiment' of each final paragraph, 'bags of words' emphasizing different attributes such as natural talent ('ability') or perseverance ('grindstone') were created (e.g., [Trix and Psenka, 2003](#); [Schmader et al., 2007](#)). This resulting categorization was validated by a questionnaire sent to academic economists based in UK research-intensive universities.

We assign the reference letters' final paragraphs (henceforth 'reference letters' for brevity) to mutually exclusive 'letter types'. We do so by classifying the letters depending on the most emphasized attributes. This discrete classification differs from the continuous outcome studied in [Eberhardt et al. \(2023\)](#) and informs the design of our experiments, while also enabling closer comparison of results. Letters classified as *ability* (respectively, *grindstone*) type are those containing a frequency of 'ability' ('grindstone') attributes above the full sample median tf-idf. This naturally leads to two further types: a *neutral* type with letters containing neither attribute and a *mixed* type with letters containing both attributes.

In addition to the reference letter data, we use information on both candidates (e.g., gender; ethnicity; PhD-granting institution; publication record; research field; initial placement outcome) and recommenders (e.g., gender; institutional affiliation and RePEc ranking; seniority; time since their PhD award; number of reference letters in the sample they wrote).

### A.4.2 Descriptives

Table [A.9](#) provides sample statistics for our letter analysis in the JM data. The unit of observation here is the reference letter (n=8,624).

---

<sup>23</sup>Our observational study received ethical clearance by the Nottingham School of Economics Research Ethics Committee in March 2020. Confidential data was handled with the approval of the University of Nottingham ethics board and in accordance with all University of Nottingham regulations, as well as an agreement signed on April 13, 2020, between the university and the job application platform through which the data were collected.

<sup>24</sup>In order to match the experimental setting, we limit the analysis to candidates who either graduated less than a year before submitting their application or will graduate in the Summer after the job market. They make up the vast majority of candidates in the sample.

<sup>25</sup>[Baltrunaite et al. \(2022\)](#) use embedding methods to analyze letters, obtaining very similar results.

Table A.9: Summary Statistics — *JM* data

|                                                           | Mean    | SD     | N    |
|-----------------------------------------------------------|---------|--------|------|
| <i>Candidate:</i>                                         |         |        |      |
| Female                                                    | 0.30    | 0.46   | 8624 |
| Ethnic White                                              | 0.42    | 0.49   | 8624 |
| Ethnic Asian                                              | 0.33    | 0.47   | 8624 |
| Ethnic Black                                              | 0.02    | 0.13   | 8624 |
| Ethnic American Indian                                    | 0.00    | 0.06   | 8624 |
| Ethnic Hispanic                                           | 0.10    | 0.30   | 8624 |
| dto withheld                                              | 0.13    | 0.33   | 8624 |
| Years since PhD                                           | 0.10    | 0.30   | 8624 |
| Publication count                                         | 0.68    | 1.38   | 8624 |
| Top 5 count                                               | 0.01    | 0.09   | 8624 |
| Top Field Journal count                                   | 0.02    | 0.16   | 8624 |
| Top General Interest Journal count                        | 0.01    | 0.09   | 8624 |
| Single-Authored JMP                                       | 0.80    | 0.40   | 8624 |
| Persons acknowledged in JMP                               | 20.87   | 18.63  | 8624 |
| Organisations acknowledged in JMP                         | 8.40    | 11.06  | 8624 |
| JMP acknowledgements missing                              | 0.05    | 0.22   | 8624 |
| PhD Institution top-25 (RePEc)                            | 0.21    | 0.40   | 8624 |
| PhD Institution top-26 to 50 (RePEc)                      | 0.15    | 0.35   | 8624 |
| PhD Institution top-51 to 100 (RePEc)                     | 0.15    | 0.36   | 8624 |
| PhD Institution top-101 to 200 (RePEc)                    | 0.19    | 0.40   | 8624 |
| PhD Institution top-201 to 500 (RePEc)                    | 0.21    | 0.41   | 8624 |
| PhD Institution outside top-500 or not listed (RePEc)     | 0.09    | 0.28   | 8624 |
| <i>Letter writer:</i>                                     |         |        |      |
| Female                                                    | 0.15    | 0.36   | 8624 |
| Institution top-25 (RePEc)                                | 0.20    | 0.40   | 8624 |
| Institution top-26 to 50 (RePEc)                          | 0.12    | 0.33   | 8624 |
| Institution top-51 to 100 (RePEc)                         | 0.16    | 0.37   | 8624 |
| Institution top-101 to 200 (RePEc)                        | 0.17    | 0.37   | 8624 |
| Institution top-201 to 500 (RePEc)                        | 0.18    | 0.38   | 8624 |
| Institution outside top-500 or not listed (RePEc)         | 0.17    | 0.38   | 8624 |
| Assistant Professor                                       | 0.09    | 0.29   | 8624 |
| Associate Professor                                       | 0.18    | 0.39   | 8624 |
| Full Professor/Chair                                      | 0.70    | 0.46   | 8624 |
| Other position                                            | 0.02    | 0.15   | 8624 |
| Years since PhD                                           | 19.91   | 11.10  | 8624 |
| Total number of letters in our sample                     | 4.10    | 3.36   | 8624 |
| <i>Letter:</i>                                            |         |        |      |
| Total word count                                          | 1121.79 | 550.02 | 8624 |
| Positive signal ("All departments hiring...")             | 0.26    | 0.44   | 8624 |
| Negative signal ("All departments outside the top-20...") | 0.13    | 0.34   | 8624 |
| Comparison with past JM candidates                        | 0.06    | 0.24   | 8624 |



### A.4.3 Analysis

Table A.10 presents results from multinomial logit regressions of letter choices on candidate gender and other control variables. Column 1 reports the full sample results (marginal effects), columns 2 to 9 the split sample results by recommender gender, institution (top-100 or not), academic rank (associate and full professor or not) and PhD cohort (before 2004 or not). These results are illustrated in the upper panel of Figure 8 in the main text.

Table A.10: Letter choice (multinomial logit) — Heterogeneity by writer characteristics

| Writer                                                                        | (1)<br>All         | (2)<br>Gender<br>M | (3)<br>F        | (4)<br>Top-100 Institution<br>Yes | (5)<br>Institution<br>No | (6)<br>Assoc/Full Prof<br>Yes | (7)<br>Prof<br>No | (8)<br>PhD Cohort<br>Early | (9)<br>Late     |
|-------------------------------------------------------------------------------|--------------------|--------------------|-----------------|-----------------------------------|--------------------------|-------------------------------|-------------------|----------------------------|-----------------|
| <i>Relative Propensity of a Female Candidate to receive reference type...</i> |                    |                    |                 |                                   |                          |                               |                   |                            |                 |
| Ability                                                                       | -3.53***<br>(1.07) | -4.02***<br>(1.18) | -1.19<br>(2.38) | -4.27***<br>(1.59)                | -3.10**<br>(1.43)        | -3.78***<br>(1.13)            | -0.56<br>(3.04)   | -4.97***<br>(1.39)         | -1.24<br>(1.61) |
| Grindstone                                                                    | 2.53***<br>(0.95)  | 2.37**<br>(1.02)   | 3.14<br>(2.47)  | 3.62***<br>(1.33)                 | 1.75<br>(1.32)           | 2.76***<br>(0.99)             | 0.50<br>(2.98)    | 4.44***<br>(1.19)          | -0.35<br>(1.57) |
| Mixed                                                                         | 1.42<br>(1.05)     | 1.31<br>(1.15)     | 1.47<br>(2.60)  | 1.96<br>(1.52)                    | 0.97<br>(1.46)           | 1.88*<br>(1.11)               | -2.47<br>(3.10)   | 0.34<br>(1.35)             | 2.83*<br>(1.65) |
| Neutral                                                                       | -0.43<br>(1.13)    | 0.35<br>(1.24)     | -3.42<br>(2.47) | -1.31<br>(1.65)                   | 0.39<br>(1.56)           | -0.86<br>(1.23)               | 2.53<br>(3.08)    | 0.19<br>(1.53)             | -1.23<br>(1.68) |
| Letters                                                                       | 8624               | 7340               | 1284            | 4159                              | 4465                     | 7644                          | 980               | 5060                       | 3564            |
| Candidates                                                                    | 2881               | 2833               | 1056            | 1753                              | 1965                     | 2857                          | 851               | 2624                       | 2241            |
| Writers                                                                       | 4257               | 3527               | 730             | 1877                              | 2504                     | 3590                          | 667               | 2258                       | 1999            |
| Pseudo R <sup>2</sup>                                                         | 0.01               | 0.01               | 0.05            | 0.02                              | 0.01                     | 0.01                          | 0.06              | 0.01                       | 0.02            |
| Controls                                                                      | Yes                | Yes                | Yes             | Yes                               | Yes                      | Yes                           | Yes               | Yes                        | Yes             |

*Note:* We conduct heterogeneity analysis by splitting the sample by writer characteristics. All models include year dummies and full controls for candidate (including about their JMP), recommender (years since PhD, rank, gender, institutional rank), and letter characteristics (including predictions/signals about their placement prospects and other sentiments expressed in the letters). Results from the multinomial logit models are predicted probabilities (standard errors are clustered at the candidate level), the letter types are ability, grindstone, both, and neither (without any ordering imposed). Letter types (dichotomous) are determined by adopting the median tf-idf value by candidate for ability and grindstone as cut-off. We exclude candidates who have graduated more than one year ago at the time of application as well as recommenders who are not in academia. Columns (2) and (3) split the sample by recommender gender, (4) and (5) distinguishes recommender institutions in the top 100, (6) and (7) by academic rank (assistant prof vs. associate/full professor), and (8) and (9) by the year they were awarded their PhD (before 2004 is labelled as 'Early').

Table A.11 again presents results from multinomial logit regressions of letter choices on candidate gender and other control variables, but based on the sample splits of recommenders predicted to be gender informed or uninformed. Predictions are based on the analysis of gender awareness and demographic characteristics in the academic survey data, with marginal effects of individual characteristics then applied to the JM data to create a ‘propensity score’. Column 1 of Table A.11 presents the full sample results for all recommenders we could include in the prediction exercise, columns 2 and 3 for the sample split of a predicted propensity of 0.5 (which is also the median split). The results in columns 2 and 3 of the table are illustrated in the lower panel of Figure 8 in the main text. Columns 4 to 6 of the table explore more restrictive definitions of gender awareness.

Table A.11: Letter choice (multinomial logit) — Heterogeneity by Predicted Gender Awareness

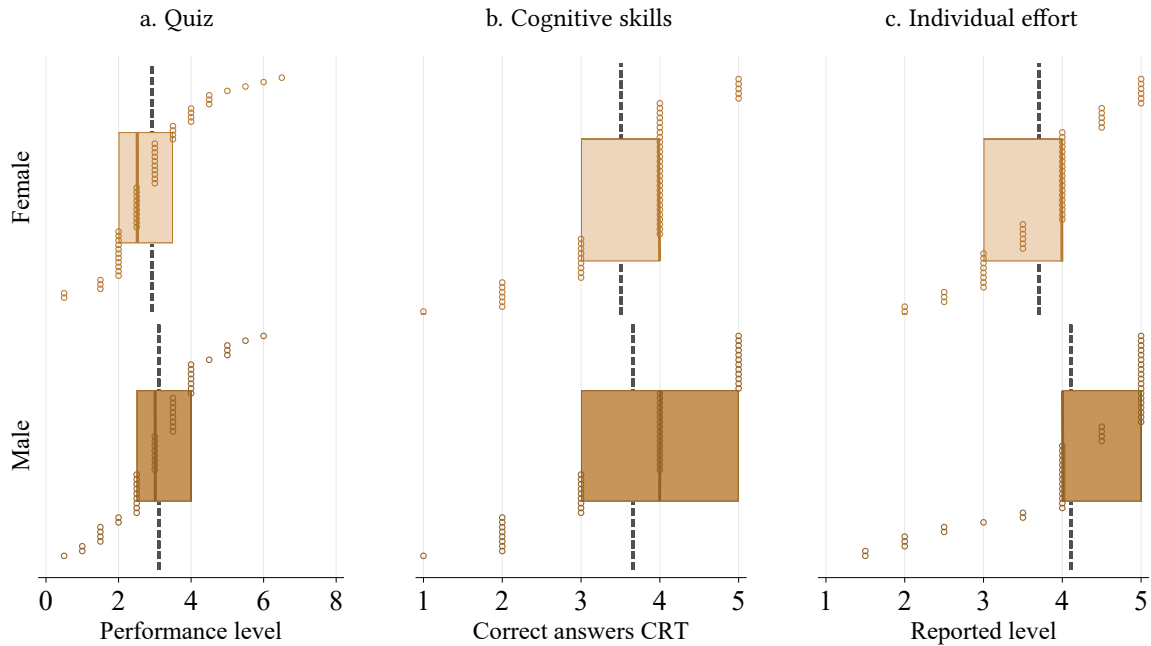
|                        | (1)                | (2)                | (3)               | (4)                | (5)             | (6)                | (7)              |
|------------------------|--------------------|--------------------|-------------------|--------------------|-----------------|--------------------|------------------|
| Threshold (Propensity) | N/A                | 0.5                |                   | 0.55               |                 | 0.6                |                  |
| Informed Views         | N/A                | No                 | Yes               | No                 | Yes             | No                 | Yes              |
| Ability letter         | -3.47***<br>(1.08) | -4.97***<br>(1.30) | -0.34<br>(1.85)   | -4.14***<br>(1.20) | -0.69<br>(2.24) | -3.85***<br>(1.16) | -1.65<br>(2.78)  |
| Grindstone letter      | 2.50***<br>(0.96)  | 3.21***<br>(1.14)  | 0.94<br>(1.78)    | 2.46**<br>(1.05)   | 2.30<br>(2.29)  | 2.56**<br>(1.01)   | 1.85<br>(2.87)   |
| Mixed letter           | 1.56<br>(1.07)     | 0.52<br>(1.28)     | 3.49*<br>(1.87)   | 1.38<br>(1.17)     | 1.80<br>(2.36)  | 1.05<br>(1.14)     | 5.35*<br>(3.02)  |
| Neutral letter         | -0.59<br>(1.15)    | 1.23<br>(1.41)     | -4.09**<br>(1.91) | 0.30<br>(1.29)     | -3.41<br>(2.32) | 0.24<br>(1.23)     | -5.54*<br>(2.87) |
| Letters                | 8427               | 5774               | 2653              | 6877               | 1550            | 7497               | 930              |
| Candidates             | 2873               | 2699               | 1738              | 2810               | 1225            | 2842               | 805              |
| Writers                | 4102               | 2765               | 1378              | 3235               | 883             | 3582               | 528              |
| Pseudo R <sup>2</sup>  | 0.01               | 0.01               | 0.02              | 0.01               | 0.04            | 0.01               | 0.06             |
| Controls               | Yes                | Yes                | Yes               | Yes                | Yes             | Yes                | Yes              |

*Note:* We present split sample results using different cutoffs of the predicted propensity of gender awareness (based on the linear probability model in the academic survey data): a predicted value of 0.5, 0.55, or 0.6, in each case splitting the sample into recommenders predicted to hold or not hold informed views as indicated. All models include full controls for candidate (including about their JMP), recommender (years since PhD, rank, institutional rank), and letter characteristics (including predictions/signals about their placement prospects and other sentiments expressed in the letters). Results from the multinomial logit models are predicted probabilities (standard errors are clustered at the candidate level), the letter types are ability, grindstone, both, and neither (without any ordering imposed). Letter types (dichotomous) are determined by adopting the median tf-idf value by candidate for ability and grindstone as cut-off. We exclude candidates who have graduated more than one year ago at the time of application as well as recommenders from non-academic institutions and those from countries which were not represented among participants in the academic survey.

## A.5 Are male and female candidates different?

Figure A.3 displays comparisons between female and male candidates in the online experiment (N=100). The figure compares the average values and distributions for female and male candidates across three different metrics: quiz performance (panel a), cognitive skills (panel b) and individual effort (panel c). We find no difference in either the average Quiz performance (Wilcoxon rank-sum test,  $z = 1.17$ ,  $p = 0.25$ ) or measured cognitive skills (ibid.,  $z = 0.74$ ,  $p = 0.46$ ).<sup>26</sup> The only dimension on which female and male candidates differ is average self-reported effort, with men claiming significantly higher levels (ibid.,  $z = 2.74$ ,  $p = 0.006$ ). Thus, if anything, men should be more likely to receive a grindstone letter. Finally, if we restrict the analysis to workers with a performance of 5, 6, or 7 correct answers in Quiz A, all differences are insignificant.

Figure A.3: Comparison of female and male candidates



*Note:* Quiz performance level measures how many questions (out of 10) were solved correctly by the candidate. Cognitive ability is measured by the number of correctly answered questions in a Cognitive Reflection Test (CRT) ranging from 0 to 5. Effort is self-reported on a scale from 1 to 5. Small circles represent individual responses, dashed lines means, thick lines medians and boxes the interquartile range.

Figures A.4 and A.5 show expected candidate qualities for recommenders and recruiters respectively in the online experiment. Recommenders observe a candidate's performance in Quiz A when forming their assessment, whereas recruiters only observe the text of the letter (and candidate gender). We find no significant differences between male and female candidates. If anything, female candidates of the same quality as their male counterparts (same Quiz A performance, same letter) are seen as having slightly higher ability and effort. This stresses that differences in letter or hiring choices are not driven by candidates being different or being perceived as different.

<sup>26</sup>This is also true when looking at Quiz A and Quiz B performance separately.

Figure A.4: Expected candidate qualities among recommenders in the online experiment

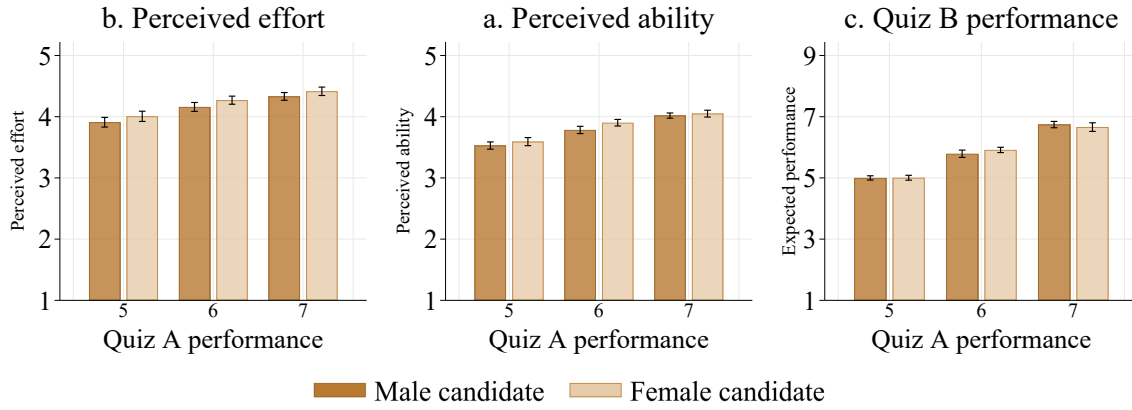
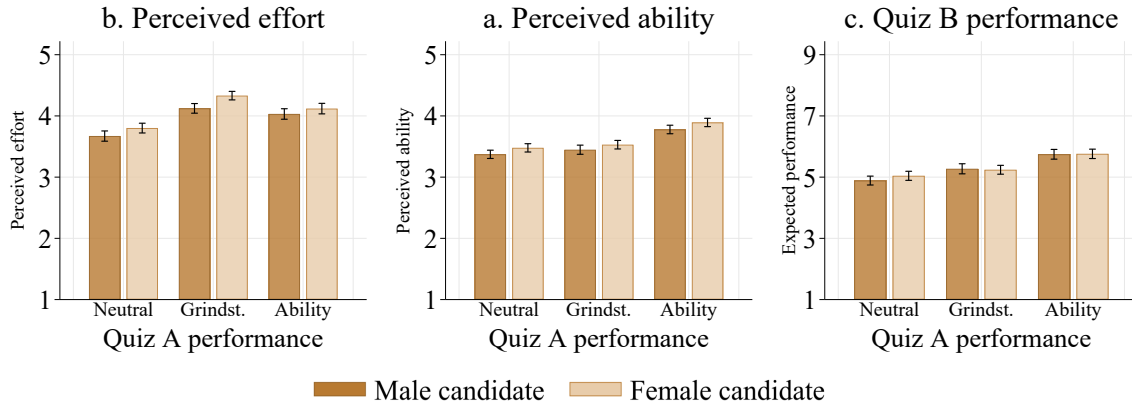


Figure A.5: Expected candidate qualities among recruiters in the online experiment



## A.6 Strategic Considerations and Beliefs

Table A.12 shows results for regressing letter choice on candidate gender in the academic survey while controlling for beliefs about the best letter for a candidate. Table A.13 does the same for the online experiment. We see that any significant gender differences disappear after controlling for beliefs, highlighting the crucial role of strategic considerations. These results are represented in Figure 10 of the main text.

Table A.12: Letter choices and beliefs in the academic survey - weighted multinomial logits

|                               | <b>Full sample</b> |                 | <b>Informed</b>  |                 | <b>Uninformed</b>  |                 |
|-------------------------------|--------------------|-----------------|------------------|-----------------|--------------------|-----------------|
|                               | (1)                | (2)             | (3)              | (4)             | (5)                | (6)             |
| Controlling for beliefs       | No                 | Yes             | No               | Yes             | No                 | Yes             |
| Ability - Female candidate    | -3.05<br>(6.04)    | 2.62<br>(4.99)  | 13.73*<br>(8.15) | 7.61<br>(6.36)  | -14.61**<br>(7.34) | -3.36<br>(6.80) |
| Grindstone - Female candidate | -2.05<br>(3.65)    | -1.67<br>(3.29) | -4.83<br>(4.17)  | -3.52<br>(3.97) | -0.02<br>(5.01)    | 0.10<br>(4.58)  |
| Neutral - Female candidate    | 5.10<br>(5.65)     | -0.95<br>(4.42) | -8.90<br>(7.70)  | -4.09<br>(5.84) | 14.63**<br>(6.93)  | 3.26<br>(6.13)  |
| Observations                  | 505                | 503             | 231              | 230             | 274                | 273             |
| Referrer controls             | Yes                | Yes             | Yes              | Yes             | Yes                | Yes             |
| Pseudo R <sup>2</sup>         | 0.04               | 0.31            | 0.12             | 0.39            | 0.08               | 0.32            |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Beliefs control for which letter recommenders believe to be preferred by most recruiters. It is a categorical variable that takes the value 0 if the neutral letter is believed to be preferred by most, 1 if it is the grindstone, and 2 if it is the ability letter. Recommender controls include gender, academic position, institutional rank, letter writing experience, and hiring experience. All regressions control for order effects.

Table A.13: Letter choices and beliefs in the online experiment - multinomial logits

|                               | <b>Full sample</b> |                 | <b>Non-stereotyped</b> |                 | <b>Stereotyped</b> |                 |
|-------------------------------|--------------------|-----------------|------------------------|-----------------|--------------------|-----------------|
|                               | (1)                | (2)             | (3)                    | (4)             | (5)                | (6)             |
| Controlling for beliefs       | No                 | Yes             | No                     | Yes             | No                 | Yes             |
| Ability - Female candidate    | 0.41<br>(3.37)     | 0.46<br>(3.17)  | 7.65**<br>(3.86)       | 2.04<br>(3.73)  | -14.94**<br>(6.21) | -6.27<br>(6.90) |
| Grindstone - Female candidate | 2.84<br>(2.85)     | 0.55<br>(2.71)  | 1.57<br>(3.52)         | 2.37<br>(3.34)  | 4.57<br>(4.55)     | 0.22<br>(4.76)  |
| Neutral - Female candidate    | -3.25<br>(3.49)    | -1.01<br>(3.40) | -9.22**<br>(4.05)      | -4.41<br>(3.94) | 10.37*<br>(6.10)   | 6.05<br>(6.78)  |
| Observations                  | 769                | 769             | 555                    | 555             | 214                | 214             |
| Referrer controls             | Yes                | Yes             | Yes                    | Yes             | Yes                | Yes             |
| Pseudo R <sup>2</sup>         | 0.06               | 0.15            | 0.06                   | 0.18            | 0.14               | 0.16            |

\* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01. Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Beliefs control for which letter recommenders believe to be preferred by most recruiters. It is a categorical variable that takes the value 0 if the neutral letter is believed to be preferred by most, 1 if it is the grindstone, and 2 if it is the ability letter. Recommender controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. Columns (3) and (4) do not control for education due to insufficient variation for estimation. All regressions control for order effects and the candidate's performance in Quiz A.

Table A.14 shows results for the online sample when using weights. Results are qualitatively similar to Table A.13. The only difference is that in the sub-sample with gendered views, women are still significantly less likely to receive an ability letter, even after controlling for beliefs. This is again in line with the stronger effects we uncover on hiring and letter choices when re-weighting the online

sample.

Table A.14: Letter choices and beliefs in the online experiment - weighted multinomial logits

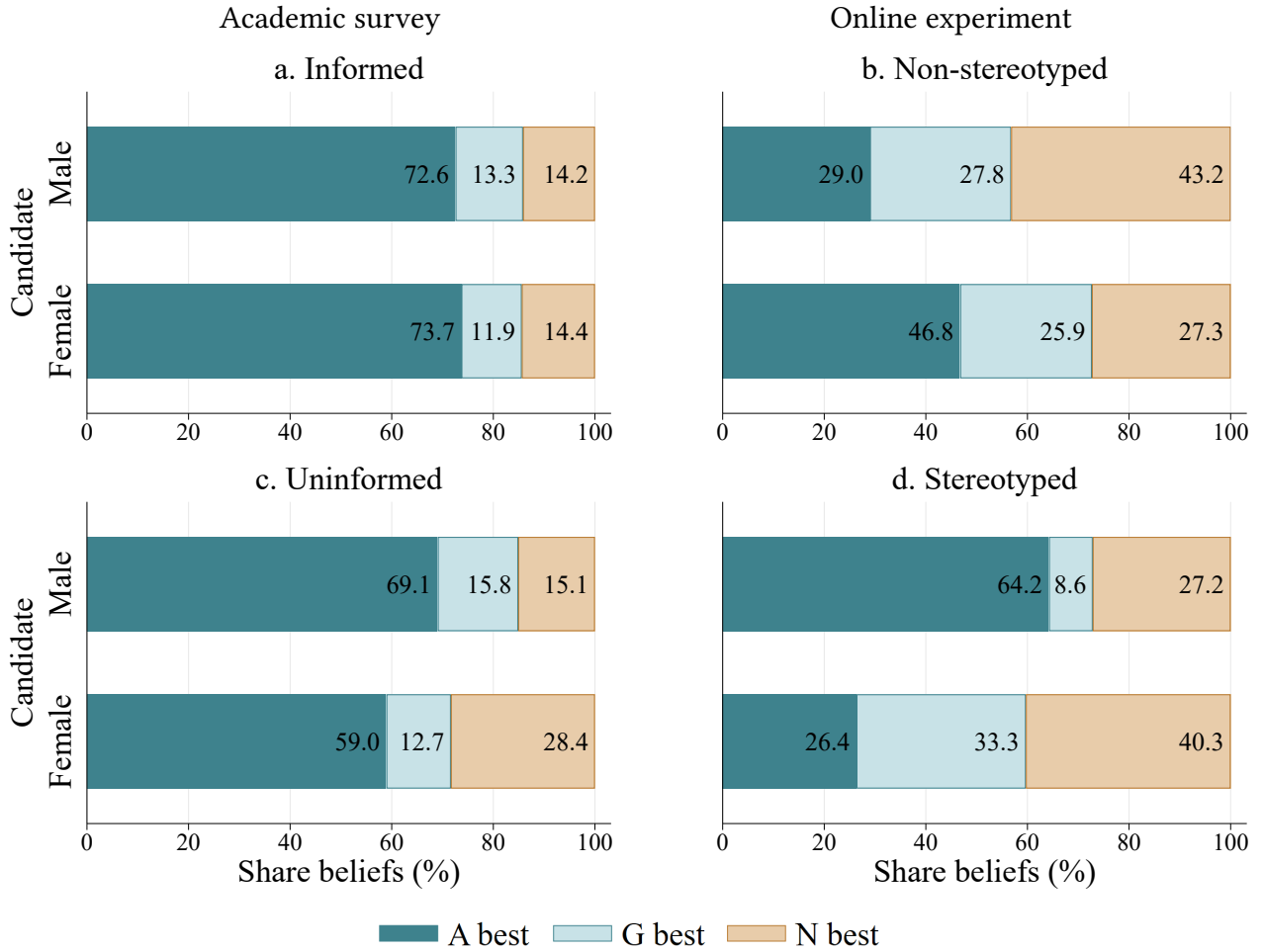
|                               | <b>Full sample</b> |                 | <b>Non-stereotyped</b> |                 | <b>Stereotyped</b>  |                   |
|-------------------------------|--------------------|-----------------|------------------------|-----------------|---------------------|-------------------|
|                               | (1)                | (2)             | (3)                    | (4)             | (5)                 | (6)               |
| Controlling for beliefs       | Yes                | No              | Yes                    | No              | Yes                 | No                |
| Ability - Female candidate    | 0.22<br>(4.26)     | -1.07<br>(3.98) | 8.42*<br>(4.76)        | 3.58<br>(4.61)  | -25.79***<br>(7.11) | -15.24*<br>(8.24) |
| Grindstone - Female candidate | 2.67<br>(3.25)     | 1.15<br>(3.08)  | 1.76<br>(3.91)         | 1.94<br>(3.59)  | 7.95*<br>(4.78)     | 2.29<br>(4.63)    |
| Neutral - Female candidate    | -2.89<br>(4.28)    | -0.08<br>(4.04) | -10.18**<br>(4.80)     | -5.52<br>(4.50) | 17.84**<br>(7.34)   | 12.95<br>(8.02)   |
| Observations                  | 769                | 769             | 555                    | 555             | 214                 | 214               |
| Referrer controls             | Yes                | Yes             | Yes                    | Yes             | Yes                 | Yes               |
| Pseudo R <sup>2</sup>         | 0.06               | 0.17            | 0.07                   | 0.21            | 0.16                | 0.21              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Robust standard errors in parentheses.

*Note:* The dependent variable is 0 if a neutral letter was chosen, 1 if a grindstone letter was chosen, and 2 if an ability letter was chosen. Recommender controls include age, gender, education level, nationality, strategic thinking ability, risk aversion, perceived candidate quality, and perceived required skills to perform well in the quiz. Column (3) does not control for education due to insufficient variation for estimation. Sampling weights are calculated using the observational sample as the underlying population. All regressions control for order effects and the candidate's performance in Quiz A.

Figure A.6 decomposes the beliefs about which letter maximizes a candidate's probability to be hired by gender views. In both the academic survey and the online experiment, we find that participants with uninformed or stereotyped views are much more likely to believe that the ability letter is best for male candidates than for female candidates. In the online experiment, we observe the opposite pattern for people with non-stereotyped views, while participants with informed views in the academic survey have similar beliefs independent of the candidate's gender.

Figure A.6: Beliefs about the most effective letter by gender views



*Note:* Share of recommenders believing a given letter maximizes the candidate's probability to be hired. In reality, the best letter for a candidate (independent of gender) is the ability one.

## A.7 Discrimination Decomposition

### Gender views/Stereotyping

Table A.15 shows the same decomposition as Table 1 in the main text. In addition to calculating the average direct discrimination (ADD) based on the distribution of letters in the overall sample, the table also reports results from using the distribution of letters for female and male candidates instead. Results are qualitatively the same, independently of which distribution is used.

Table A.16 uses weighted regressions for the decomposition in the online experiment. Again, results are qualitatively similar to the unweighted results, albeit more pronounced.

Table A.15: Decomposition of total into systemic and direct discrimination - Gender views

|                                | <i>Academic survey</i> |                       |       | <i>Online experiment</i> |                         |       |
|--------------------------------|------------------------|-----------------------|-------|--------------------------|-------------------------|-------|
|                                | (1)                    | (2)                   | (3)   | (4)                      | (5)                     | (6)   |
|                                | Full sample            | Informed views<br>Yes | No    | Full sample              | Stereotyped views<br>No | Yes   |
| <b>Total discrimination</b>    |                        |                       |       |                          |                         |       |
| TD                             | 1.55                   | 8.08                  | -4.70 | 1.34                     | 4.13                    | -5.21 |
| <b>Direct discrimination</b>   |                        |                       |       |                          |                         |       |
| ADD ( $p(s_k)$ sample)         | 3.52                   | 3.86                  | 2.95  | 0.38                     | 0.10                    | 0.99  |
| ADD ( $p(s_k)$ female)         | 3.47                   | 4.37                  | 2.35  | 0.38                     | 0.43                    | 0.18  |
| ADD ( $p(s_k)$ male)           | 3.57                   | 3.38                  | 3.51  | 0.37                     | -0.26                   | 1.67  |
| <b>Systemic discrimination</b> |                        |                       |       |                          |                         |       |
| SD ( $p(s_k)$ sample)          | -1.97                  | 4.22                  | -7.65 | 0.96                     | 4.03                    | -6.19 |
| SD ( $p(s_k)$ female)          | -1.92                  | 3.71                  | -7.05 | 0.96                     | 3.70                    | -5.39 |
| SD ( $p(s_k)$ male)            | -2.02                  | 4.71                  | -8.21 | 0.97                     | 4.40                    | -6.88 |
| Recommenders                   | 506                    | 232                   | 274   | 959                      | 679                     | 280   |
| Recruiters                     | 514                    | 514                   | 514   | 959                      | 959                     | 959   |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column.



Table A.16: Decomposition of total into systemic and direct discrimination - Gender views, weighted online sample

|                                | <i>Academic survey</i> |                    |       | <i>Online experiment</i> |                      |        |
|--------------------------------|------------------------|--------------------|-------|--------------------------|----------------------|--------|
|                                | (1)                    | (2)                | (3)   | (4)                      | (5)                  | (6)    |
|                                | Full sample            | Informed views Yes | No    | Full sample              | Stereotyped views No | Yes    |
| <b>Total discrimination</b>    |                        |                    |       |                          |                      |        |
| TD                             | 1.55                   | 8.08               | -4.70 | 2.67                     | 6.10                 | -6.32  |
| <b>Direct discrimination</b>   |                        |                    |       |                          |                      |        |
| ADD ( $p(s_k)$ sample)         | 3.52                   | 3.86               | 2.95  | 1.26                     | 0.93                 | 2.17   |
| ADD ( $p(s_k)$ female)         | 3.47                   | 4.37               | 2.35  | 1.39                     | 1.82                 | 0.21   |
| ADD ( $p(s_k)$ male)           | 3.57                   | 3.38               | 3.51  | 1.12                     | -0.04                | 3.97   |
| <b>Systemic discrimination</b> |                        |                    |       |                          |                      |        |
| SD ( $p(s_k)$ sample)          | -1.97                  | 4.22               | -7.65 | 1.41                     | 5.18                 | -8.50  |
| SD ( $p(s_k)$ female)          | -1.92                  | 3.71               | -7.05 | 1.28                     | 4.28                 | -6.53  |
| SD ( $p(s_k)$ male)            | -2.02                  | 4.71               | -8.21 | 1.54                     | 6.14                 | -10.29 |
| Recommenders                   | 506                    | 232                | 274   | 959                      | 679                  | 280    |
| Recruiters                     | 514                    | 514                | 514   | 959                      | 959                  | 959    |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column.

## Recruiter sub-samples

Table A.17 shows decomposition results when distinguishing between *recruiters* with different gender views. As can be seen there are no systematic differences between recruiters depending on their gender views. This underscores that it is the behavior of recommenders and not that of recruiters driving our results.

Table A.17: Decomposition of total into systemic and direct discrimination - recruiter sub-samples

|                                | <i>Academic survey</i> |                       |       | <i>Online experiment</i> |                         |       |
|--------------------------------|------------------------|-----------------------|-------|--------------------------|-------------------------|-------|
|                                | (1)                    | (2)                   | (3)   | (4)                      | (5)                     | (6)   |
|                                | Full sample            | Informed views<br>Yes | No    | Full sample              | Stereotyped views<br>No | Yes   |
| <b>Total discrimination</b>    | 1.55                   | 1.43                  | 2.51  | 1.34                     | 2.85                    | 0.93  |
| <b>Direct discrimination</b>   |                        |                       |       |                          |                         |       |
| ADD ( $p(s_k)$ sample)         | 3.52                   | 3.96                  | 3.73  | 0.38                     | 1.91                    | -0.08 |
| ADD ( $p(s_k)$ female)         | 3.47                   | 4.19                  | 3.37  | 0.38                     | 1.67                    | 0.01  |
| ADD ( $p(s_k)$ male)           | 3.57                   | 3.75                  | 4.07  | 0.37                     | 2.16                    | -0.17 |
| <b>Systemic discrimination</b> |                        |                       |       |                          |                         |       |
| SD ( $p(s_k)$ sample)          | -1.97                  | -2.54                 | -1.22 | 0.96                     | 0.94                    | 1.02  |
| SD ( $p(s_k)$ female)          | -1.92                  | -2.77                 | -0.86 | 0.96                     | 1.19                    | 0.93  |
| SD ( $p(s_k)$ male)            | -2.02                  | -2.32                 | -1.56 | 0.97                     | 0.69                    | 1.11  |
| Recommenders                   | 506                    | 506                   | 506   | 959                      | 959                     | 959   |
| Recruiters                     | 514                    | 257                   | 257   | 959                      | 706                     | 253   |

*Note:* The table reports average estimates across 1,000 simulations per column.

## Beliefs

Table A.18 shows decomposition results for the sub-samples of participants with correct or incorrect beliefs. This is analogous to Table 2 in the main text. However, here we additionally report results when calculating ADD based on the distribution of letters among male and female candidates. The findings are qualitatively very similar to using the distribution in the overall sample. Table A.19 reports the same analysis with weighted regressions for the online sample.

Table A.18: Decomposition of total into systemic and direct discrimination - Beliefs

|                                | <i>Academic survey</i> |                        |       | <i>Online experiment</i> |                        |       |
|--------------------------------|------------------------|------------------------|-------|--------------------------|------------------------|-------|
|                                | (1)                    | (2)                    | (3)   | (4)                      | (5)                    | (6)   |
|                                | Full sample            | Correct beliefs<br>Yes | No    | Full sample              | Correct beliefs<br>Yes | No    |
| <b>Total discrimination</b>    |                        |                        |       |                          |                        |       |
| TD                             | 1.55                   | 8.40                   | -7.92 | 1.34                     | 2.35                   | 0.36  |
| <b>Direct discrimination</b>   |                        |                        |       |                          |                        |       |
| ADD ( $p(s_k)$ sample)         | 3.52                   | 5.05                   | 0.07  | 0.38                     | 2.11                   | -0.83 |
| ADD ( $p(s_k)$ female)         | 3.47                   | 5.37                   | -0.16 | 0.38                     | 2.13                   | -0.87 |
| ADD ( $p(s_k)$ male)           | 3.57                   | 4.77                   | 0.34  | 0.37                     | 2.09                   | -0.80 |
| <b>Systemic discrimination</b> |                        |                        |       |                          |                        |       |
| SD ( $p(s_k)$ sample)          | -1.97                  | 3.36                   | -7.98 | 0.96                     | 0.24                   | 1.20  |
| SD ( $p(s_k)$ female)          | -1.92                  | 3.03                   | -7.76 | 0.96                     | 0.22                   | 1.23  |
| SD ( $p(s_k)$ male)            | -2.02                  | 3.63                   | -8.26 | 0.97                     | 0.27                   | 1.16  |
| Recommenders                   | 506                    | 344                    | 160   | 959                      | 391                    | 568   |
| Recruiters                     | 514                    | 514                    | 514   | 959                      | 959                    | 959   |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column. Correct beliefs restricts the analysis to recommenders who correctly believe that the ability letter will be most successful. In the academic sample, two recommenders did not answer the beliefs question. The table presents results using the distribution of letters for all candidates.

Table A.19: Decomposition of total into systemic and direct discrimination - Beliefs, weighted online sample

|                                | <i>Academic survey</i> |                        |       | <i>Online experiment</i> |                        |       |
|--------------------------------|------------------------|------------------------|-------|--------------------------|------------------------|-------|
|                                | (1)                    | (2)                    | (3)   | (4)                      | (5)                    | (6)   |
|                                | Full sample            | Correct beliefs<br>Yes | No    | Full sample              | Correct beliefs<br>Yes | No    |
| <b>Total discrimination</b>    |                        |                        |       |                          |                        |       |
| TD                             | 1.55                   | 8.40                   | -7.92 | 2.67                     | 6.45                   | -1.20 |
| <b>Direct discrimination</b>   |                        |                        |       |                          |                        |       |
| ADD ( $p(s_k)$ sample)         | 3.52                   | 5.05                   | 0.07  | 1.26                     | 4.46                   | -1.38 |
| ADD ( $p(s_k)$ female)         | 3.47                   | 5.37                   | -0.16 | 1.39                     | 4.88                   | -1.67 |
| ADD ( $p(s_k)$ male)           | 3.57                   | 4.77                   | 0.34  | 1.12                     | 3.99                   | -1.09 |
| <b>Systemic discrimination</b> |                        |                        |       |                          |                        |       |
| SD ( $p(s_k)$ sample)          | -1.97                  | 3.36                   | -7.98 | 1.41                     | 1.99                   | 0.18  |
| SD ( $p(s_k)$ female)          | -1.92                  | 3.03                   | -7.76 | 1.28                     | 1.57                   | 0.48  |
| SD ( $p(s_k)$ male)            | -2.02                  | 3.63                   | -8.26 | 1.54                     | 2.45                   | -0.10 |
| Recommenders                   | 506                    | 344                    | 160   | 959                      | 391                    | 568   |
| Recruiters                     | 514                    | 514                    | 514   | 959                      | 959                    | 959   |

*Note:* The table reports average differences in hiring probability for female versus male candidates across 1,000 simulations per column. Correct beliefs restricts the analysis to recommenders who correctly believe that the ability letter will be most successful. In the academic sample, two recommenders did not answer the beliefs question. The table presents results using the distribution of letters for all candidates.

## B Instructions & Procedures

### B.1 Sample construction academic survey

Invitations to the academic survey were sent out in two waves to the email addresses of academics that we collected from publicly available websites.

For the first wave, we used email addresses of academics in the RePEc top 5% institutions (universities and colleges) based in the following countries: Australia, Belgium, Canada, Denmark, France, Germany, Hong Kong, Ireland, Israel, Italy, Netherlands, New Zealand, Norway, PR China, Singapore, South Korea, Spain, Sweden, Switzerland, the United States, and the United Kingdom. We excluded international organizations, central banks, and research-only or policy institutions.

Our goal was to reach an overall sample size of one thousand survey participants. As this was not reached in our first wave, we invited more academics to participate in a second wave. For this we extended our study population to the top 8% institutions in the US, UK, Ireland, Canada, and Australia, because institutions in these countries are the most likely to engage regularly with the economics job market.

### B.2 Instructions Academic survey

Below we provide the full instructions for the academic survey and the online experiment. We added notes (in italics) to provide more details on the survey flow and to explain which questions were randomized.

#### B.2.1 Consent

Thanks a lot for participating in this study!

#### Study information

The aim of this study is to develop a better understanding of the decision making of academics in the recruitment process for junior faculty members. In the following, we will show you a short biography of a **hypothetical candidate along with the final paragraph of three different reference letters**. We will then ask you some questions about how you perceive the final paragraphs.

The study should take **5-7 minutes**. Your participation is crucial for our research, and we are very grateful for your support. As a thank-you token, we donate **£2 to UNICEF** for every completed survey. In addition, for completing this survey you can win one of fifty **£50-Amazon vouchers**. To be considered for the vouchers, you may leave your email address when prompted at the end of the survey. By completing this survey you receive one ticket to win a voucher. Throughout the study you will have the chance to receive more tickets depending on your responses. Winners will

be chosen at random after all responses have been collected and will be contacted by the end of July. Similarly, if you would like to be informed about the **results of our study** once it is completed, you may indicate this at the end. Your email address will only be used for these purposes.

#### Contact information

If you have any questions, concerns or complaints about this research, its procedures, risks and benefits, contact us at [cedex@nottingham.ac.uk](mailto:cedex@nottingham.ac.uk).

#### Consent

Please note that you are free to withdraw at any point while completing the questionnaire. All data collected will be kept confidential and used for research purposes only.

By clicking the **‘I consent’ button**, you agree to the following:

1. I have read and understood the information about the study.
2. I confirm that I have been given enough information about this study, and I voluntarily agree to take part.

If you do not agree, please click the **‘I don’t consent’** button. Thank you for your time.

- I consent
- I do not consent

#### B.2.2 Recommender survey

*Note: The following questions were only presented to half of the participants who were randomly assigned to act as recommenders.*

#### Application - NAME Miller

*Note: Half of the participants are randomized to see Julia instead of NAME, the other half to see Thomas*

Imagine you agreed to write a reference letter for NAME Miller, a PhD student who goes on the economics job market. In the box below we provide a very **short description** of this hypothetical student:

NAME Miller is a fifth-year PhD student in Economics with a focus on applied micro. NAME’s PhD institution is ranked among the top 20 institutions globally. NAME’s job market paper exploits

a large and quasi-randomized 4 year-long income tax holiday to identify intertemporal labor responses to taxation among high-wage earners. It is likely to be published in a top general interest journal or at least in a top field journal.

We will now show you the final paragraph of three reference letters. This final paragraph can be seen as a summary and a conclusion of the letter. **Your task is to choose your preferred paragraph for this student's letter.** By preferred we mean, which paragraph is most in line with how you would end a reference letter for this student. You will be given the option to read the full letter, although this is not necessary for the study.

*Note: There was a button participants could click to read the full letter. The full letter is provided in Section [B.3](#) below. It is identical across the three letter types save for the final paragraph.*

### **Reference letter - NAME Miller**

1) Imagine you were asked to write a reference letter for NAME. Which of the three final paragraphs would you choose?

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

*Note: The order of the three final paragraphs is randomized across participants. Passages were highlighted in bold as indicated below.*

### Letter 1 - final paragraph

NAME is a remarkable job market applicant who **has the talent to come up with interesting, innovative, and policy-relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that HE/SHE **has the technical skills and versatility to push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA'd for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is an **extremely bright and creative deep thinker**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the ability letter.*

### Letter 2 - final paragraph

NAME is a remarkable job market applicant who **has always been keen to put in the effort needed to come up with interesting, innovative, and policy-relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that SHE/HE **has the drive and stamina to push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA'd for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is an **extremely hard-working and dedicated researcher**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the grindstone letter.*



### Letter 3 - final paragraph

NAME is a remarkable job market applicant who **has come up with interesting, innovative and policy relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that HE/SHE **can push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA'd for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is a **great candidate**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the neutral letter.*

### Hiring prospects for NAME Miller - Own evaluation

2) Now imagine that you yourself received an application for a candidate with NAME's profile, how likely would you be to invite NAME for an interview if the application was supported by...

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

3) Among the three final paragraphs, which one convinces you most to invite NAME to an interview?

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

### Hiring prospects for NAME Miller - Views of other participants

4) We show the selected final paragraphs together with NAME's profile to **other participants** in the study who are all academic economists. Among all participants, what do you think was the most frequently chosen answer to the question: "How likely would you be to invite NAME for an

interview?” after participants read...

For each answer that you guess correctly you will receive 5 additional tickets to win one of the £50 Amazon vouchers.

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

5) What do you think most **other participants** answered to the following question: “Among the three final paragraphs, which one convinces you most to invite NAME to an interview?”

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

### B.2.3 Recruiter Survey

#### Application - NAME Miller

*Note: Half of the participants are randomized to see Julia instead of NAME, the other half to see Thomas*

Imagine you are recruiting on the economics job market. In the box below we provide a very **short description** of this hypothetical student:

NAME Miller is a fifth-year PhD student in Economics with a focus on applied micro. NAME's PhD institution is ranked among the top 20 institutions globally. NAME's job market paper exploits a large and quasi-randomized 4 year-long income tax holiday to identify intertemporal labor responses to taxation among high-wage earners. It is likely to be published in a top general interest journal or at least in a top field journal.

In addition, you received a reference letter for this candidate. We will now show you the final paragraph of the letter. This final paragraph can be seen as a summary and a conclusion of the letter. **Your task is to decide whether you want to invite NAME for an interview.** Although this is not necessary for the study, you may click on the link below the paragraph to read the full letter.

*Note: There was a button participants could click to read the full letter. The full letter is available in Section [B.3](#) below. It is identical across letter types besides the final paragraph.*

#### Letter 1 - NAME Miller

*Note: The order of the three final paragraphs is randomized across participants.*

##### Letter 1 - final paragraph

NAME is a remarkable job market applicant who **has the talent to come up with interesting, innovative, and policy-relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that HE/SHE **has the technical skills and versatility to push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA'd for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is an **extremely bright and creative deep thinker**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the ability letter.*

1) How likely would you be to invite NAME for an interview?

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

2) We ask the same question to **other participants** in this study who are all academic economists. Out of all participants, what do you think was the most frequent answer to the question: “How likely would you be to invite NAME for an interview?”

If you guess correctly you will receive 5 additional tickets to win one of the £50 Amazon vouchers.

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

## **Letter 2 - NAME Miller**

Imagine the same candidate received a different recommendation letter with the following final paragraph. Although this is not necessary for the study, you again have the option to read the full letter.

As a reminder we also show you again NAME’s profile:

NAME Miller is a fifth-year PhD student in Economics with a focus on applied micro. NAME’s PhD institution is ranked among the top 20 institutions globally. NAME’s job market paper exploits a large and quasi-randomized 4 year-long income tax holiday to identify intertemporal labor responses to taxation among high-wage earners. It is likely to be published in a top general interest journal or at least in a top field journal.

### **Letter 2 - final paragraph**

NAME is a remarkable job market applicant who **has always been keen to put in the effort needed to come up with interesting, innovative, and policy-relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that SHE/HE **has the drive and stamina to push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA’d for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is an **extremely hard-working and dedicated researcher**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the grindstone letter.*

3) How likely would you be to invite NAME for an interview?

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

4) We ask the same question to **other participants** in this study who are all academic economists. Out of all participants, what do you think was the most frequent answer to the question: “How likely would you be to invite NAME for an interview?”

If you guess correctly you will receive 5 additional tickets to win one of the £50 Amazon vouchers.

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

### Letter 3 - NAME Miller

Finally, imagine the final paragraph of NAME’s reference letter reads as shown below.

As a reminder we also show you again NAME’s profile:

NAME Miller is a fifth-year PhD student in Economics with a focus on applied micro. NAME’s PhD institution is ranked among the top 20 institutions globally. NAME’s job market paper exploits a large and quasi-randomized 4 year-long income tax holiday to identify intertemporal labor responses to taxation among high-wage earners. It is likely to be published in a top general interest journal or at least in a top field journal.

#### Letter 3 - final paragraph

NAME is a remarkable job market applicant who **has come up with interesting, innovative and policy relevant ideas**. This is well reflected in HIS/HER completed projects. Furthermore, NAME has a very promising and unusually developed research agenda. NAME has shown that HE/SHE **can push HIS/HER projects until completion**. I expect NAME to continue generating good research ideas and have a productive career. As for HIS/HER classroom performance, HE/SHE has already proven to be a great teacher. NAME received excellent student evaluations for the courses HE/SHE TA’d for me. Moreover, NAME is also a pleasant and collegial person who is a pleasure to interact with. In sum, NAME is a **great candidate**. All departments should consider HIS/HER application seriously and carefully. I recommend NAME enthusiastically.

*Note: This is the neutral letter.*

5) How likely would you be to invite NAME for an interview?

*Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

6) We ask the same question to **other participants** in this study who are all academic economists. Out of all participants, what do you think was the most frequent answer to the question: “How likely would you be to invite NAME for an interview?”

If you guess correctly you will receive 5 additional tickets to win one of the £50 Amazon vouchers. *Note: Answers for each letter are presented on a Likert scale from 0 (definitely not) to 10 (definitely invite).*

7) Among the three final paragraphs, which one convinces you most to invite NAME to an interview?

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

8) We ask the same question to **other participants** in this study who are all academic economists. What do you think most other participants answered to the following question: “Among the three final paragraphs, which one convinces you most to invite NAME to an interview?”

If you guess correctly you will again receive 5 additional tickets to win one of the £50 Amazon vouchers. You can have another look at the final paragraphs by clicking on the corresponding button.

- Letter 1 - final paragraph
- Letter 2 - final paragraph
- Letter 3 - final paragraph

### B.2.4 Questionnaire (common to recommender and recruiter surveys)

Thank you very much! You completed the main part of the survey. Before you go, we would like to ask you a few last questions about yourself and your hiring experience.

1) When considering whether to invite a job market candidate for an interview, how important are the following characteristics to you?

Please rank them from most to least important by **dragging the options** into your preferred order.

Most important

- Intellectual rigour
- Technical skills
- Hard-working
- Teaching experience
- Quality of the job market paper
- Rank of the PhD institution
- Social skills

Least important

2) Your hiring experience

a. How often have you been part of a junior hiring committee?

- Never
- 1-5
- 6-10
- 11-20
- >20

b. How many reference letters have you already written for PhD students (approximately)?

- Never
- 1-5

- 6-10
- 11-20
- >20

### 3) About this survey

a. How familiar are you with the literature on gender bias?

*Note: Answers on a Likert scale from 0 (very unfamiliar) to 5 (very familiar).*

b. Have you read any of the following papers?

- Dupas et al. (2021). “Gender and the Dynamics of Economics Seminars”
- Koffi (2021). “Innovative ideas and gender inequality”
- Hengel (2022). “Are Women Held to Higher Standards? Evidence from Peer Review”
- Wu (2020). “Gender Bias among Professionals: An Identity-Based Interpretation”
- Eberhardt et al. (2022). “Gender Differences in Reference Letters: Evidence from the Economics Job Market”

*Note: Order of papers is randomized. Answer options are "Yes" and "No".*

### 4) Lottery

a. Would you like to participate in the draw for the Amazon vouchers?

- Yes
- No

b. Would you like to be informed of the results of this study?

- Yes
- No

In case you answered yes to either question 4a or 4b, please enter your **email address** here:

---



Thank you very much again for your participation!

If you have any comments, you can leave them below. Any feedback/ additional thoughts regarding this study are highly appreciated.

---

**Please click finish to submit your answers.**

### **B.3 Full reference letter**

In the study participants could click a link to read the full reference letter of the candidate. The letter was identical in all treatments except for the last paragraph as outlined above. The full letter can be found below:

\*\*\*\*\*

To whom it may concern

Letter of Recommendation for NAME Miller

This letter is to recommend NAME, a fifth year graduate student at my university, for a position in your department. NAME is an applied-micro researcher, working on taxation in the context of country YYY. I have been one of NAME's advisors over the past four years. Hence, I know NAME well. NAME has produced a very interesting job market paper, convincingly identifying the effects of income taxation on high earning employees using a unique natural experiment carried out in YYY. NAME has been very productive and has two other completed papers on taxation, which NAME will be able to submit very soon.

NAME's job market paper exploits a large and quasi-randomized 4 year-long income tax holiday to identify intertemporal labor responses to taxation among high-wage earners. Eligibility for the tax holiday was based on whether past (and thus pre-determined) wage earnings were below a fixed threshold, creating a discontinuity that affected workers active in the same labor market with sharply different marginal tax rates. Using rich population-wide administrative data and a regression discontinuity design, NAME estimates a precise and large elasticity of earnings with respect to the marginal tax rate.

Interestingly, NAME finds that behavioural responses are larger for more flexible outcomes (overtime hours) and more elastic subgroups of the population. Another important result uncovered by NAME is that a significant number of self-employed workers who regularly worked for the same firms switched their status to permanent salaried workers for their previous clients. This result highlights the importance of tax incentives to determine the nature of work and is suggestive of co-operation between workers and employers in determining the type of contract they choose.

To my knowledge, this is one of the best identified papers on the short-term effects of taxes on labor supply. The additional results indicating that employer and employee cooperation is crucial to

deliver tax avoidance are also in line with a set of very recent studies that have focused on payroll taxes. As a result, I am highly confident that this paper will end up published in a top general interest/top field journal. Importantly, I believe that it will become a reference for future studies on the real effects of income taxation on high wage income earners. Because of the simplicity and clarity of the context, this paper could also become a widely used teaching reference, both at the advanced undergraduate and PhD levels.

NAME has 2 additional completed drafts, again focusing on country YYY.

In the paper “Evidence from a change in the payment system of means tested transfers” NAME and a fellow PhD student show that the way in which child benefits are disbursed affects employer’s behavior and female labor supply. To tackle this question, they take advantage of a change in the payment system that was gradually rolled out between 1995 and 2001 in YYY. Under the old system, employers were intermediaries that paid child benefits to their employees together with their salary and were allowed to deduct this transfer from employer social security contributions. The new system centralized instead the payments in a government agency that started disbursing the allowances directly to eligible workers. Using detailed employer-employee data, NAME and his/her co-author show that employers reduce wages after the reform, disproportionately affecting female labor supply. The effect is mostly driven by big firms and more generally by firms that enjoy a significant degree of monopsonistic power in the labor market. This paper contributes to the public economics literature showing that employee vs. employer administration, even when economically neutral, can end up having a large impact in practice.

In NAME’s last paper “Taxpayers’ responses to tax changes”, NAME estimates the response of self-employed and firms to a new turnover tax, using rich administrative data. In particular, NAME exploits several revenue-dependent discontinuities that provide incentives to underreport taxable income combined with a bunching design to estimate revenue elasticities. Consistent with the literature, this paper uncovers tax avoidance as a response to such opportunities.

\*\*\*\*\*

This constitutes the main body of the letter. It is followed by the last paragraph. The last paragraph is either grindstone, ability or neutral and shown in the instructions of the academic survey.

## **B.4 Instructions Online experiment**

### **B.4.1 Workers**

#### **Welcome to this study!**

**Study information** In this study, you will be asked to attempt two quizzes (A and B) and fill out a short survey. Each quiz has 10 questions from different subjects: math, business, and history. In other words, you will answer 20 questions in total. All questions for a quiz will be displayed on a single page and you can answer questions in the order you want to. You will have **2 minutes to work on each quiz**.

**How much can I earn?** You will be paid £1.80 for your participation. In addition, you can earn an additional bonus of up to £1.50 depending on your answers. For your bonus payment, one of the two quizzes will be randomly selected and you will get **£0.10 for every correct answer in that quiz**. It is in your best interest to treat each quiz as if it determines your bonus payment. You can win another £0.50 in the survey following the quizzes.

**How long will it take?** The study will take around 7 minutes to complete.

**Do I have to participate?** No, you are under no obligation to take part. You are free to withdraw at any point before or while completing the study. All data collected will be kept confidential and used for research purposes only.

### **Consent**

Please note that you are free to withdraw at any point while completing the questionnaire. All data collected will be kept confidential and used for research purposes only.

By clicking the **‘I consent’** button, **you agree to the following:**

1. I have read and understood the information about the study.
2. I understand that my responses will remain anonymous.
3. I confirm that I have been given enough information about this study, and I voluntarily agree to take part.

If you do not agree, please click the **‘I don’t consent’** button. Thank you for your time.

- I consent
- I don’t consent

In online studies, there are sometimes bots or participants who do not read the instructions. This means that there are random answers which compromise the results of research studies. To show that you read our questions carefully, please enter ‘Turquoise’ as your answer to the next question. Based on the text above, what is your favourite colour?

- Blue
- Green

- Turquoise
- Purple
- Red

ID Please enter your Prolific ID below:

---

Recall that you have **2 minutes** to work on each Quiz. Once you click continue the timer for Quiz A will start.

### Quiz A

*Note: The order of Quiz A and Quiz B was randomized between participants.*

Q1) How many original colonies were there in the US?

- Eleven
- Twelve
- Thirteen
- Fourteen

Q2) What kingdom is the oldest monarchy in Europe?

- United Kingdom
- Denmark
- Germany
- Sweden

Q3) What city was the US Declaration of Independence signed?

- Washington DC
- New York City
- Philadelphia
- Boston

Q4) Giovanni wants to buy shirts that cost £19.40 each and sweaters that cost £24.80 each. An 8% sales tax will be applied to the entire purchase. If Giovanni buys 2 shirts, which equation relates the number of sweaters purchased,  $p$ , and the total cost in dollars,  $y$ ?

- $1.08(38.80 + 24.80p) = y$
- $38.80 + 24.80p = 0.92y$
- $38.80 + 24.80p = 1.08y$
- $0.92(38.80 + 24.80p) = y$

Q5) The equation  $y=36+18x$  models the relationship between the height  $y$ , in inches, of a typical golden delicious apple tree and the number of years,  $x$ , after it was planted. If the equation is graphed in the  $xy$ -plane, what is indicated by the  $y$ -intercept of the graph?

- The age, in years, of a typical apple tree when it is planted
- The height, in inches, of a typical apple tree when it is planted
- The number of years it takes a typical apple tree to grow
- The number of inches a typical apple tree grows each year

Q6)  $-3x-4y = 20$

$x-10y = 16$

If  $(x,y)$  is the solution to the system of equations above, what is the value of  $x$ ?

- -14
- -12
- -4
- 16

Q7) Which of the following is an example of £-cost averaging?

- Index funds
- 401(k) plans
- Certificates of deposit

Q8) Suppose you have £100 in a savings account earning 2 percent interest a year. After five years,

how much would you have?

- More than £102
- Exactly £102
- Less than £102

Q9) If interest rates rise, what will typically happen to bond prices? Rise, fall, stay the same, or is there no relationship?

- Rise
- Fall
- Stay the same
- No relationship

Q10) Buying a single company's stock usually provides a safer return than a stock mutual fund.

- True
- False

Thank you, you have finished Quiz A!

How much effort did you put into doing Quiz A on a scale from 1 to 5?

*Note: Answers for each letter are presented on a Likert scale from 1 (little effort) to 5 (a lot of effort).*

Once you click continue, the **2 minute timer** for Quiz B will start.

## Quiz B

Q1) Who is the only British prime minister to have been assassinated?

- William Pitt the Elder
- Spencer Perceval
- George Canning

Q2) Which infamous incident of treachery in Scotland is said to have inspired the extremely bloody “Red Wedding” massacre scene in the TV series Game of Thrones?

- The Black Dinner of 1440
- The Glencoe Massacre of 1692
- The murder of Lord Darnley in 1567

Q3) In the immediate aftermath of the Boston Massacre, many of the citizens of the city of Boston:

- decided to enlist in the British army in order to obtain weapons to defend themselves.
- organized the Boston Tea Party.
- circulated propaganda in the form of pamphlets and prints.
- looked to George Washington to organize and lead a new colonial military force.

Q4) A line is graphed in the xy-plane. If the line has a positive slope and a negative y-intercept, which of the following points cannot lie on the line?

- (-3, -3)
- (-3, 3)
- (3, -3)
- (3, 3)

Q5) A parachute design uses 18 separate pieces of rope. Each piece of rope must be at least 270 centimeters and no more than 280 centimeters long. What inequality represents all possible values of the total length of rope  $x$ , in centimeters, needed for the parachute?

- $270 \leq x \leq 280$
- $4,860 \leq x \leq 4,870$
- $4,860 \leq x \leq 5,040$
- $5,030 \leq x \leq 5,040$

Q6)  $(x^2y^2)^{(1/2)}(x^2y^3)^{(1/3)} = x^{(a/3)}y^{(a/2)}$

If the equation above, where  $a$  is a constant, is true for all positive values of  $x$  and  $y$ , what is the value of  $a$ ?

- 2
- 3
- 5
- 6

Q7) If the equation  $y=(x-6)(x+12)$  is graphed in the  $xy$ -plane, what is the  $x$ -coordinate of the parabola's vertex?

- -6
- -3
- 3
- 6

Q8) Which of the following orders of the four investment instruments according to their average volatility (from 1 = low to 4 = high) is correct?

- 1. Savings account 2. Government bonds 3. Stocks 4. Corporate bonds
- 1. Savings account 2. Government bonds 3. Corporate bonds 4. Stocks
- 1. Government bonds 2. Savings account 3. Stocks 4. Corporate bonds
- 1. Government bonds 2. Savings account 3. Corporate bonds 4. Stocks

Q9) Which one of the following statements is NOT a possible advantage of investing in investment funds from the perspective of an investor?

- The possibility to invest diversified
- The possibility to invest in special markets
- The possibility to invest small amounts of money
- The possibility to participate in the choice of individual stocks

Q10) Which one of the following terms is synonymous with keeping a sell option?

- Short Put
- Long Put



- Short Call
- Long Call

Thank you, you have finished Quiz B!

How much effort did you put into doing Quiz B on a scale from 1 to 5?

*Note: Answers for each letter are presented on a Likert scale from 1 (little effort) to 5 (a lot of effort).*

Thank you for finishing the main part of the experiment!

Before you leave, we will ask you 5 last questions to test your reasoning ability. For each question you answer correctly, you will earn an additional £0.10.

1) If you're running a race and you pass the person in second place, what place are you in?

---

2) A farmer had 15 sheep and all but 8 died. How many are left?

---

3) Emily's father has three daughters. The first two are named April and May. What is the third daughter's name?

---

4) How many cubic feet of sand are there in a hole that is 3' deep x 3' wide x 3' long?

---

5) An expedition on a mountain climbing trip was traveling with eleven horse packs. Each horse can carry only three packs. How many horses does the expedition need?

---

Finally, we would like to get some information about yourself.

How old are you?

---

What is your gender?

- Male
- Female
- Non-binary/ third gender
- Prefer not to say

What is the highest level of education you completed?

- No formal education
- High school or equivalent
- College/ undergraduate degree
- Master's degree/ MBA
- Doctoral Degree (PhD)

Thank you very much again for your participation! Any feedback/ additional thoughts regarding this study is highly appreciated. If you have any comments, you can leave them below:

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**Please click finish to submit your answers.**

#### **B.4.2 Recommenders**

**Welcome to this study!**

**Study information** In this study we will ask you to evaluate and predict the performance of workers on an online crowdsourcing platform. In addition, you will get the opportunity to recommend a worker to a recruiter.

**How much can I earn?** You will be paid £1.50 for your participation. In addition, you can earn an additional bonus of up to £1.50 depending on your choices and the choices of other participants.

**How long will it take?** The study will take around 10 minutes to complete.

**Do I have to participate?** No, you are under no obligation to take part. You are free to withdraw at any point before or while completing the study. All data collected will be kept confidential and

used for research purposes only.

### **Consent**

Please note that you are free to withdraw at any point while completing the questionnaire. All data collected will be kept confidential and used for research purposes only.

By clicking the '**I consent**' button, **you agree to the following**:

1. I have read and understood the information about the study.
2. I understand that my responses will remain anonymous.
3. I confirm that I have been given enough information about this study, and I voluntarily agree to take part.

If you do not agree, please click the '**I don't consent**' button. Thank you for your time.

- I consent
- I don't consent

In online studies, there are sometimes bots or participants who do not read the instructions. This means that there are random answers which compromise the results of research studies. To show that you read our questions carefully, please enter 'Turquoise' as your answer to the next question. Based on the text above, what is your favourite colour?

- Blue
- Green
- Turquoise
- Purple
- Red

ID Please enter your Prolific ID below:

---

### **Instructions**

Today you will decide how to recommend a worker to a recruiter.

### The workers

In a previous study, we asked people to complete 2 quizzes of similar difficulty: Quiz A and B. Each quiz consisted of 10 questions about **math, finance, and history**. Workers were paid for finishing the quiz and received an additional bonus for getting randomly selected questions correct. You can see a few sample questions here:

Q7) If the equation  $y = (x-6)(x+12)$  is graphed in the  $xy$ -plane, what is the  $x$ -coordinate of the parabola's vertex?

☐ -6

☐ -3

☐ 3

☐ 6

Q8) Who is the only British prime minister to have been assassinated?

☐ William Pitt the Elder

☐ Spencer Perceval

☐ George Canning

Q9) If interest rates rise, what will typically happen to bond prices? Rise, fall, stay the same, or is there no relationship?

☐ Rise

☐ Fall

☐ Stay the same

☐ No relationship

### Your task

We will show you **Quiz A** performance and demographic information of one worker. You will then be asked to recommend the worker to a recruiter (see below), using one of three pre-written messages.

We will also ask you a few questions about how you perceive the performance of the worker. **If**

**your guess about the worker is correct, you will earn an additional bonus of £0.50.**

### **The recruiters**

**Recruiters receive your message in addition to demographic information about the worker.**

They are informed about the content of the quiz but do not know anything about the worker's performance. The recruiter will then decide whether to hire the worker.

If the worker is hired, the **recruiter receives a bonus (between £0 and £1) depending on how many questions the worker answered correctly in Quiz B.** If they do not hire the worker, they receive a fixed payment. If the worker performed above average, the recruiter earns more by hiring the worker. If the worker performed below average it is better for the recruiter not to hire the worker.

**If a recruiter hires the worker you recommended, you receive an additional bonus of £0.50.**

We will also ask you a few questions about the recruiter's behaviour. Again, **if you guess correctly, you will earn an additional £0.50.**

On the next screen, you will be asked some comprehension questions to make sure that you understand the instructions.

We are now asking you a few questions about the set-up. If you need to consult the instructions again, you can do so by clicking on the button below.

*Note: Participants could only continue once they selected the correct answer.*

1) Questions in Quiz A are:

- Harder than in Quiz B
- Easier than in Quiz B
- Comparable to Quiz B

2) To make a hiring decision the recruiter sees:

- Quiz A performance
- Quiz B performance
- Your message

3) Which of the following statements is true:

- Recruiters earn more the better a hired worker performs
- Recruiters earn more the more workers they hire
- Recruiters do not care about the worker's performance

4) You are earning an additional bonus if:

- The worker you recommend is hired
- The worker performs well in Quiz A
- The worker performs well in Quiz B

### The worker

Below you see information about a worker:

*Note: Participants randomly saw 5, 6 or 7 correctly solved questions instead of  $x$  and either male or female instead of  $y$ .*

|                                      |        |
|--------------------------------------|--------|
| Correctly solved questions in Quiz A | $x/10$ |
| Worker's gender                      | $y$    |

Note: On average workers answered 3.5/10 questions correctly in Quiz A.

We first ask you how you perceive the performance of this worker. We will randomly select one out of the questions below to determine your bonus.

**If your guess about the worker is correct, you will receive an additional £0.50.** It is in your best interest to treat each question as if it determines your bonus payment.

1) How many questions do you think did the worker shown above answer correctly in **Quiz B** (out of 10)?

*Note: Possible answers range from 0 to 10.*

2) In addition to the quizzes, we elicited the effort workers put into the task on a scale from 1 to 5. **How much effort** do you think did the worker shown above put into solving Quiz A?

*Note: Possible answers range from 1 (little effort) to 5 (a lot of effort).*

3) Finally, we tested the workers' cognitive ability. Workers could score between 1 and 5 points. How do you think did the worker shown above perform in the **cognitive ability** test?

*Note: Possible answers range from 1 (lowest ability) to 5 (highest ability).*

## About Quiz A

1) How much **effort** do you think is required to perform well (answering at least 5 questions correct) in Quiz A?

*Note: Possible answers range from 1 (little effort) to 5 (a lot of effort).*

2) How much **cognitive ability** do you think is required to perform well (answering at least 5 questions correct) in Quiz A?

*Note: Possible answers range from 1 (little cognitive ability) to 5 (a lot of cognitive ability).*

3) Recall that your worker was **y** and solved **x** questions correctly in Quiz A. How much do you think was this performance due to effort and how much due to cognitive ability (in %)?

Note that the two values you select should add up to 100%.

\_\_\_\_\_ Performance due to **effort**

\_\_\_\_\_ Performance due to **cognitive ability**

## Recommendation decision

Here is a reminder of the worker.

|                                      |      |
|--------------------------------------|------|
| Correctly solved questions in Quiz A | x/10 |
| Worker's gender                      | y    |

Note: On average workers answered 3.5/10 questions correctly in Quiz A.

## Which message do you want to send to the recruiter to recommend your worker?

Recall that recruiters will only see your recommendation message and the worker's gender to make a hiring decision.

*Note: The order of the different messages is randomized.*

- Having observed the worker's performance in Quiz A, I believe that HE/SHE will **put in the effort** to perform well in Quiz B. I think HE/SHE is **very thorough**.

- Having observed the worker's performance in Quiz A, I believe that HE/SHE has the **talent** to perform well in Quiz B. I think HE/SHE is **very bright**.
- Having observed the worker's performance in Quiz A, I believe that HE/SHE **can perform** well in Quiz B. I think HE/SHE is a **good worker**.

### About the recruiters

We now ask you some questions about the recruiters. To determine your bonus, we will randomly select one of the questions below and from the next page.

**If your guess about the recruiter is correct, you will receive an additional £0.50.** It is in your best interest to treat each question as if it determines your bonus payment.

*Note: The order of messages is randomized.*

We will ask many recruiters to make a hiring decision based on the three messages that you have seen before.

1) Out of 100 recruiters how many will on average hire a **y worker** after reading the following message?

*"Having observed the worker's performance in Quiz A, I believe that HE/SHE will **put in the effort** to perform well in Quiz B. I think HE/SHE is **very thorough**."*

*Note: Answer possibilities go from 0 to 100 in steps of 10.*

2) Out of 100 recruiters how many will on average hire a **y worker** after reading the following message?

*"Having observed the worker's performance in Quiz A, I believe that HE/SHE has the **talent** to perform well in Quiz B. I think HE/SHE is **very bright**."*

*Note: Answer possibilities go from 0 to 100 in steps of 10.*

3) Out of 100 recruiters how many will on average hire a **y worker** after reading the following message?

*"Having observed the worker's performance in Quiz A, I believe that HE/SHE **can perform** well in Quiz B. I think HE/SHE is a **good worker**."*



*Note: Answer possibilities go from 0 to 100 in steps of 10.*

4) We will also show recruiters three y candidates with different messages, and ask them to hire one of them. **Which candidate do you think will get hired?**

The candidate with the message...

*Note: The order of messages is randomized.*

- Having observed the worker's performance in Quiz A, I believe that HE/SHE will **put in the effort** to perform well in Quiz B. I think HE/SHE is **very thorough**.
- Having observed the worker's performance in Quiz A, I believe that HE/SHE has the **talent** to perform well in Quiz B. I think HE/SHE is **very bright**.
- Having observed the worker's performance in Quiz A, I believe that HE/SHE **can perform** well in Quiz B. I think HE/SHE is a **good worker**.

5) Finally, we will ask recruiters whether y workers are characterized more by talent or effort. **What do you think most of them answered?**

- y workers are characterized more by **talent**
- y workers are characterized more by **effort**
- y workers are equally characterized by **effort and talent**

### **B.4.3 Recruiters**

**Welcome to this study!**

**Study information** In this study we will ask you to predict the performance of workers on an online crowdsourcing platform. In addition, you will get the opportunity to hire one worker and receive an additional bonus based on their performance.

**How much can I earn?** You will be paid £1.00 for your participation. In addition, you can earn an additional bonus depending on your responses and the responses of other participants.

**How long will it take?** The study will take around 10 minutes to complete.

**Do I have to participate?** No, you are under no obligation to take part. You are free to withdraw at any point before or while completing the study. All data collected will be kept confidential and

used for research purposes only.

### **Consent**

Please note that you are free to withdraw at any point while completing the questionnaire. All data collected will be kept confidential and used for research purposes only.

By clicking the **'I consent'** button, **you agree to the following:**

1. I have read and understood the information about the study.
2. I understand that my responses will remain anonymous.
3. I confirm that I have been given enough information about this study, and I voluntarily agree to take part.

If you do not agree, please click the **'I don't consent'** button. Thank you for your time.

- I consent
- I don't consent

In online studies, there are sometimes bots or participants who do not read the instructions. This means that there are random answers which compromise the results of research studies. To show that you read our questions carefully, please enter 'Turquoise' as your answer to the next question. Based on the text above, what is your favourite colour?

- Blue
- Green
- Turquoise
- Purple
- Red

ID Please enter your Prolific ID below:

---

### **Instructions**

Today you will decide whether to hire workers.

### The workers

In a previous study, we asked people to complete 2 quizzes of similar difficulty: Quiz A and B. Each quiz consisted of 10 questions about **math, finance, and history**. Workers were paid for finishing the quiz and received an additional bonus for getting randomly selected questions correct. You can see a few sample questions here:

Q7) If the equation  $y = (x-6)(x+12)$  is graphed in the  $xy$ -plane, what is the  $x$ -coordinate of the parabola's vertex?

☐ -6

☐ -3

☐ 3

☐ 6

Q8) Who is the only British prime minister to have been assassinated?

☐ William Pitt the Elder

☐ Spencer Perceval

☐ George Canning

Q9) If interest rates rise, what will typically happen to bond prices? Rise, fall, stay the same, or is there no relationship?

☐ Rise

☐ Fall

☐ Stay the same

☐ No relationship

### Recommenders

We hired other Prolific workers to act as recommenders and have a closer look at the workers. Each recommender saw demographic information and **performance in Quiz A** for one worker. They were then asked to recommend the worker using one of three pre-defined messages. If you decide to hire the worker that was recommended by the recommender, the recommender will receive a bonus of £0.50.

### Your task

We will show you **demographic information and the chosen recommendation message for one worker**. You will then have to choose whether to hire the worker or not.

If you hire the worker, you will receive a bonus of **£0.10 for each question the worker answered correctly in Quiz B**. You thus earn more the more questions were answered correctly by a hired worker. If you decide not to hire the worker, you will receive a fixed bonus of £0.40.

We will then ask you to make a second hiring decision between three workers. You will again receive **£0.10 for each question the hired worker answered correctly in Quiz B**. Finally, we will ask you a few questions about how you perceive the performance of the workers. **If your guess about the worker is correct you will receive an additional bonus of £0.50.**

On the next screen, you will be asked some comprehension questions to make sure that you understand the instructions.

We are now asking you a few questions about the set-up. If you need to consult the instructions again, you can do so by clicking on the button below.

*Note: Participants could only continue once they selected the correct answer.*

1) Questions in Quiz A are:

- Harder than in Quiz B
- Easier than in Quiz B
- Comparable to Quiz B

2) The recommender knows:

- The worker's performance in Quiz A
- The worker's performance in Quiz B
- The worker's performance in both quizzes

3) If you decide to hire a worker and that worker answered 5 out of 10 questions in Quiz B correctly, what is your additional bonus?

- £0.50
- £0.80
- £1.00

4) What is your additional bonus if you decide not to hire a worker?

- £1.00
- £0.00
- £0.80

### Hiring decision 1

Below you see information about a worker:

*Note: Participants randomly saw one of the three messages instead of z and either male or female instead of y. The three messages that were randomly assigned to z are the same messages as used in the recommender survey.*

|                       |   |
|-----------------------|---|
| recommender's message | z |
| Worker's gender       | y |

Note: On average workers answered 3.5/10 questions correctly in Quiz A.

### Do you want to hire this worker?

Recall that if you hire the worker, their performance in Quiz B will determine your bonus payment (you receive £0.10 per correct answer). If you decide not to hire the worker, you receive a fixed payment of £0.40.

- Yes, I want to hire the worker.
- No, I don't want to hire the worker.

### About the worker

Here is a reminder of the worker:

|                       |   |
|-----------------------|---|
| recommender's message | z |
| Worker's gender       | y |

We now ask you how you perceive the performance of this worker. We will randomly select one of the questions below to determine your bonus.

1) How many questions do you think did the worker shown above answer correctly in **Quiz B** (out of 10)?

*Note: Possible answers range from 0 to 10.*

2) In addition to the quizzes, we elicited the effort workers put into the task on a scale from 1 to 5. **How much effort** do you think did the worker shown above put into solving Quiz A?

*Note: Possible answers range from 1 (little effort) to 5 (a lot of effort).*

3) Finally, we tested the workers' cognitive ability. Workers could score between 1 and 5 points. How do you think did the worker shown above perform in the **cognitive ability** test?

*Note: Possible answers range from 1 (lowest ability) to 5 (highest ability).*

4) Among all workers, do you think that y workers are characterized more by high ability or high effort?

- y workers are characterized more by **talent**
- y workers are characterized more by **effort**
- y workers are equally characterized by **effort and talent**

### About the quiz

1) How much **effort** do you think is required to perform well (answering at least 5 questions correct) in the quiz?

*Note: Possible answers range from 1 (little effort) to 5 (a lot of effort).*

2) How much **cognitive ability** do you think is required to perform well (answering at least 5 questions correct) in the quiz?

*Note: Possible answers range from 1 (little cognitive ability) to 5 (a lot of cognitive ability).*

### Hiring decision 2

Below you see information about three workers:

*Note: Participants randomly saw a different messages (z1, z2, z3) for each worker. The gender of the*

worker was fixed and either male or female instead of  $y$ . The three messages that were randomly assigned to  $z1$ ,  $z2$ , and  $z3$  are the same messages as used in the recommender survey.

|                       | Worker 1 | Worker 2 | Worker 3 |
|-----------------------|----------|----------|----------|
| recommender's message | $z1$     | $z2$     | $z3$     |
| Worker's gender       | $y$      | $y$      | $y$      |

**Which of the three workers do you want to hire?**

The performance of the hired worker in Quiz B will determine your bonus payment (you receive £0.10 per correct answer).

- Worker 1
- Worker 2
- Worker 3

#### **B.4.4 Questionnaire for recommenders and recruiters**

Thank you for finishing the main part of the experiment. Before you leave, we would like to ask a few questions about yourself and your way of thinking.

1) In general, how willing are you to take risks?

Please use a scale from 0 to 10, where 0 means “completely unwilling to take risks” and a 10 means you are “very willing to take risks”.

*Note: Answers are presented on a Likert scale from 0 (completely unwilling to take risks) to 10 (very willing to take risks).*

2) We asked all participants in this study to guess a number between 0 and 100. We will then calculate the mean of all choices. The person whose guess is closest to  $2/3$  of the mean wins an **additional bonus of £5**. Which number (between 0 and 100) do you want to guess?

---

3) What do you think are the most important characteristics in a potential employee?

Please rank them by **dragging the options** into your preferred order (from most important at the top to least important at the bottom).

Most important

- Hard working
- Intelligence

- Good social skills
- Prior job experience

Least important

4) How old are you?

---

5) What is your gender?

- Male
- Female
- Non-binary/ third gender
- Prefer not to say

6) What is the highest level of education you completed?

- No formal education
- High school or equivalent
- College/ undergraduate degree
- Master's degree/ MBA
- Doctoral Degree (PhD)

Thank you very much again for your participation! Any feedback/ additional thoughts regarding this study is highly appreciated. If you have any comments, you can leave them below:

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**Please click finish to submit your answers.**